

Emergent $\mathcal{PT}$ Symmetry and Exceptional Points in a Driven Dirac Impurity

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Periodic driving of correlated impurities can generate nonequilibrium steady states, but the microscopic origin of passive non-Hermitian structures in such systems is often obscured. We show that a driven, inversion-asymmetric Dirac impurity can realize a passive $\mathcal{PT}$ -symmetric projected retarded kernel without adding gain or loss to the microscopic Hamiltonian. The construction proceeds in two steps: off-shell auxiliary angular harmonics generate a real spin-odd coherent shift, and embedding this shifted channel in an energy-dependent Dirac bath converts it into a relative spin-dependent decay rate. After subtracting common bath damping, the co-decaying impurity kernel contains the passive $\mathcal{PT}$ core $\Delta_{\text{eff}}\sigma_x + i\Gamma_{\text{PT}}\sigma_z$, while the Kramers–Kronig partner $\Delta_{\text{coh}}\sigma_z$ provides a causal detuning floor that bounds the approach to exact coalescence. We solve the interacting impurity at the slave-boson mean-field level using the gauge-invariant projected hybridization $\beta = |\beta_0||b_c|$ and the physical bath-FDR spectrum. Analytic Schrieffer–Wolff and frozen Bethe–Ansatz–kernel estimates give the near-EP screening-scale enhancement; independently, the SBMF spectra show positive spectral-weight transfer into the shifted Friedel-type spectral resonance (FSR) associated with the Kondo feature in the same drive window. The SBMF weight transfer is not used as a definition of a thermodynamic Kondo temperature, and FWHM-like linewidths are assigned only where the positive resonance is locally resolved. The result is a self-contained projected-kernel description of how impurity-localized passive- $\mathcal{PT}$ nonnormality can enhance low-energy screening diagnostics while preserving causality of the underlying Hermitian system.

I. INTRODUCTION

Driven quantum systems provide a versatile route to steady states and dynamical phases that have no equilibrium analogs. When coherent drive, dissipation, and strong correlations coexist, the resulting many-body dynamics can generate fixed points that reflect both microscopic symmetry and emergent non-Hermitian structure. Recent work has shown that minimizing a driven bosonic field within a Keldysh mean-field framework can reproduce renormalization-group (RG)-like fixed points [1–3], pointing to a deeper connection between nonequilibrium stability and non-Hermitian criticality [4, 5].

Exceptional points (EPs) are non-Hermitian degeneracies where eigenvalues and eigenvectors coalesce, playing a central role in such settings. They reshape spectral topology and enable non-reciprocal transport, sensing enhancement, and dynamical control. EPs have been linked to the non-Hermitian skin effect [6], higher-order degeneracies [7, 8], and ultrafast projective protocols in both quantum and classical experiments [9, 10]. Their influence extends beyond spectra: non-Hermitian topology underpins phenomena in nonlinear photonics [11, 12], topological acoustics [13], engineered metamaterials [14]; coherent light–matter coupling and engineered dissipation can stabilize correlated states such as topological Kondo phases [15], and nonlinearities can generate instabilities captured by dynamical mean-field theory [16, 17].

Impurity systems are particularly sensitive to anisotropy and dissipation. Boundary-induced openness can mimic potential scattering [18–20], while nonlinear

perturbations modify inelastic processes and impurity coherence. Non-Hermitian phase transitions have recently been observed in condensed-matter experiments [21], and nonlinear optical platforms provide precise control of non-Hermitian topology [22–24]. In correlated impurity models, non-Hermitian features manifest through sign reversals in RG flows and causality violations in both interacting and non-interacting settings [4, 25–27]. Non-Hermitian Kondo physics has been explored via numerical renormalization group, cold-atom platforms, and Floquet-engineered conduction bands [6, 28–31]. Integrability of the Kondo model with time-dependent couplings has been established via quantum Knizhnik–Zamolodchikov equations [32], and a $\mathcal{PT}$ -protected Kondo effect has been observed in CeCo_2P_2 [15]. Benchmark studies of the two-dimensional Hubbard model have established the regime of validity of large- N slave-boson and mean-field treatments of correlated electrons [33], directly motivating the approach employed in Sec. V.

Here we show that a driven impurity embedded in a Dirac-like bath naturally develops a passive non-Hermitian kernel once higher angular harmonics and the energy-dependent Dirac bath are treated together. The off-shell auxiliary harmonics generate a real spin-odd coherent shift. The retarded Dirac bath, whose width depends on energy and spin-orbit structure, converts this shift into a spin-dependent decay rate. After subtracting the common bath damping, the remaining co-decaying kernel contains a dynamically induced passive $\mathcal{PT}$ -symmetric core, together with a Kramers–Kronig-related coherent detuning that bounds the approach to exact coalescence. We demonstrate that the resulting finite-drive near-EP structure persists under thermal equili-

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bration and under self-consistent minimization—despite arising without any explicit non-Hermitian term in the microscopic Hamiltonian. *In contrast to earlier studies where non-Hermiticity is imposed phenomenologically or emerges through renormalization-group flow [6, 26, 27], we demonstrate a complementary microscopic route in which a strictly Hermitian, periodically driven Dirac impurity produces a passive non-Hermitian relative kernel after coarse graining and removal of the common decay.* Dirac baths provide a natural setting because spin-momentum locking, angular-momentum selection rules, and an energy-dependent spectral width make the passive $\mathcal{PT}$ channel symmetry constrained.

The paper is organized as follows. Section II defines the full Hermitian Hamiltonian. Section III performs the bilinear Gaussian coarse graining and establishes the emergent $\mathcal{PT}$ symmetry. Section IV derives the steady-state Hamiltonian and the Hadamard rotation that makes hybridization diagonal. Section V presents the slave-boson mean field and the resulting shifted resonance condition. Sections VI–VII show spectral properties, phase-selective EP diagnostics, and the fluctuation–dissipation relation (FDR). Section VIII discusses real-time dynamics in the passive co-decaying frame. Section IX presents the biorthogonal Bethe Ansatz (BA) for the frozen effective model. Section X discusses near-EP low-energy scale estimates within slave-boson mean-field (SBMF) theory. Section XI concludes.

II. MODEL AND HAMILTONIAN

The full Hermitian Hamiltonian is $H = H_{\text{bath}} + H_{\text{imp}} + H_{\text{imp-bath}} + H_O + H_{\text{hyb}}[\phi(t)]$. Each term has a distinct physical role. H_{bath} describes the two-dimensional Dirac conduction band with spin-orbit coupling. H_{imp} is the correlated Anderson impurity. $H_{\text{imp-bath}}$ is the bare hybridization between the impurity and the $m = 0$ bath channel. H_O represents auxiliary angular-momentum modes generated by the trigonal crystal anisotropy. H_{hyb} couples the physical bath to the angular harmonics and is modulated by the periodic drive. We define all symbols at first use below. To prevent notation clashes, we reserve $t_{\text{av}} = (t + t')/2$ and $\tau = t - t'$ for the Wigner center and relative times, respectively, and use T_b only for the reservoir temperature. The chiral bath index is $\eta = \pm$, whereas $\zeta_\sigma = +1$ (-1) for $\sigma = \uparrow$ ($\downarrow$). The SOC strength is λ ; β_0 is the signed frozen drive-control amplitude; and $\tilde{\beta} = |\beta_0|/|b_c|$ is its nonnegative SBF-renormalized projection. The radial momentum cutoff k_{max} and the ultraviolet energy cutoff D_{uv} are conceptually distinct; in the dimensionless plots both are set numerically to $\pi/4$ as a convention. We denote the analytic bare-block EP by $\beta_{0,\text{bare}}^{\text{EP}} = \gamma = 0.50$ and the full projected numerical value by $\beta_{0,\text{proj}}^{\text{EP}} = 0.4975$.

A. Bath

We consider a two-dimensional (2D) host crystal with trigonal (C_3) anisotropy. The conduction electrons form a Dirac bath with spin-orbit coupling (SOC):

$$H_{\text{bath}} = \sum_{k,\sigma} \epsilon_k c_{k\sigma}^\dagger c_{k\sigma} + \lambda \sum_k k (c_{k\uparrow}^\dagger c_{k\downarrow} + \text{h.c.}), \quad (1)$$

where $c_{k\sigma}^\dagger$ creates a bath electron with radial momentum k (in units of inverse lattice constant) and spin $\sigma \in \{\uparrow, \downarrow\}$; $\epsilon_k = \alpha_b k^2$ ($\alpha_b > 0$), with radial cutoff $0 \leq k \leq k_{\text{max}} = \pi/4$; and $\lambda > 0$ is the spin-orbit coupling. The separate energy cutoff D_{uv} enters only logarithmic and scaling formulas below. The chiral rotation $c_{k\pm} = (c_{k\uparrow} \pm c_{k\downarrow})/\sqrt{2}$ diagonalizes H_{bath} exactly, yielding SOC-split chiral dispersions $\varepsilon_{k\pm} = \alpha_b k^2 \pm \lambda k$ with $\varepsilon_{k+} \neq \varepsilon_{k-}$ for all $k \neq 0$. Near the Dirac point the bath density of states is $\rho_{\text{bath}}(\omega) \sim |\omega|/\lambda^2$ [4], which is the origin of the pseudogap structure explored in Secs. V–VI.

B. Impurity

The correlated impurity is described by the Anderson Hamiltonian:

$$H_{\text{imp}} = \sum_\sigma \epsilon_\xi d_\sigma^\dagger d_\sigma + U n_\uparrow n_\downarrow, \quad (2)$$

where $d_\sigma^\dagger$ creates an impurity electron with spin σ , $n_\sigma = d_\sigma^\dagger d_\sigma$ is the occupation, ϵ_ξ is the bare impurity level before the projected bath-induced detuning is included, and $U > 0$ is the on-site Coulomb repulsion.

C. Impurity–Bath Hybridization

The impurity couples directly to the $m = 0$ (s-wave) bath channel via

$$H_{\text{imp-bath}} = \sum_{k,\sigma} \left(W_k c_{k,0,\sigma}^\dagger d_\sigma + \text{h.c.} \right), \quad (3)$$

where W_k is the bare hybridization amplitude. This term defines the physical coupling between the impurity and the conduction bath before the angular-momentum harmonics are introduced.

In the presence of the driven angular-momentum sector [Eqs. (4)–(5)], the $m = 0$ bath propagator becomes dressed by virtual processes involving the auxiliary modes $m \neq 0$: the auxiliary modes generate a retarded self-energy $\Sigma_\sigma^{\text{bath}}$ on the $m = 0$ bath channel (Sec. III). The impurity self-energy then arises from this dressed bath propagator through Eq. (3): it is the auxiliary dressing of the bath propagator, communicated to the impurity via the bare coupling W_k , that produces the effective non-Hermitian self-energy on the impurity sector. This two-step mechanism—auxiliary modes dress the bath, bath dresses the impurity—is made explicit in Sec. III and Supplemental Material.

D. Auxiliary Angular-Momentum Modes

The trigonal anisotropy $\beta_3(k)k^3 \cos 3\theta$ of the host crystal (where $\beta_3(k)$ is the momentum-dependent trigonal crystal-field strength, distinct from the hybridization amplitude β_0 introduced below; and θ is the in-plane momentum angle) couples the $m = 0$ (s-wave) bath channel to higher angular-momentum harmonics $m = \pm 3, \pm 6, \dots$ via the C_3 selection rule $\Delta m = \pm 3$. These higher harmonics are represented by auxiliary fermions $O_{k,m,\sigma}$ with angular momentum index $m \neq 0$:

$$H_O = \sum_{k,m \neq 0, \sigma} \epsilon_O^{(m)} O_{k,m,\sigma}^\dagger O_{k,m,\sigma} + \sum_{k,m,\sigma} \beta_3(k)k^3 O_{k,m+3,\sigma}^\dagger O_{k,m,\sigma} + \text{h.c.}, \quad (4)$$

where the mode energies satisfy $\epsilon_O^{(m)} = \epsilon_O^{(-m)}$ (inversion symmetry of the crystal). The fields $O_{k,m,\sigma}$ should be understood as a Feshbach, or Loewdin-partitioned, representation of remote higher-angular-harmonic crystal-field channels rather than as an additional low-energy bath. Their energies include the local orbital or crystal-field splitting of these nonresonant harmonics relative to the active $m = 0$ Dirac channel. In the low-energy limit $\epsilon_O^{(m)} \gg D_{uv}$, the auxiliary modes are therefore far off-shell and can be integrated out exactly within the quadratic sector; their main effect is a principal-value self-energy dressing the $m = 0$ bath propagator, which in turn induces a self-energy in the impurity sector through the bare hybridization Eq. (3) (see Sec. III and Supplemental Material). If an auxiliary harmonic approaches the conduction-band window, the same formalism gives an ordinary on-shell bath damping contribution; this spin-even damping is not identified with the passive relative $\mathcal{PT}$ channel.

E. Periodic Drive and Hybridization

The periodic drive couples the physical bath ($m = 0$) to the remote higher angular harmonics. We write the Hermitian bath-auxiliary hybridization as

$$H_{\text{hyb}}[\phi(t)] = \sum_{k,\theta,\sigma,m \neq 0} [V_m(k)e^{i3m\theta} e^{i\zeta_\sigma \phi(t)} \times c_{k,\theta,\sigma}^\dagger O_{k,m,\sigma} + \text{h.c.}]. \quad (5)$$

Here $\zeta_\uparrow = +1$ and $\zeta_\downarrow = -1$ label the spin-helicity channel selected by the spin-orbit-split Dirac bath, while $\phi(t) = (\pi/4) \sin(\Omega t)$ is the slow drive phase. In the passive mechanism the frozen control is the instantaneous angular velocity $\dot{\phi}(t_{\text{av}}) = (\pi\Omega/4) \cos(\Omega t_{\text{av}})$; the numerical parameter β_0 used below should be read as the corresponding effective frozen-kernel amplitude, not as the value of the phase itself. Hermiticity fixes the second

vertex to carry the complex-conjugate phase. Therefore the normal Gaussian contraction of a number-conserving auxiliary sector contains phase differences, not phase sums. This is important: the off-shell auxiliary sector does not by itself generate anti-Hermitian gain or loss.

Equivalently, the phase can be removed from Eq. (5) by the time-dependent unitary rotation

$$O_{k,m,\sigma} \rightarrow e^{-i\zeta_\sigma \phi(t)} O_{k,m,\sigma}. \quad (6)$$

The vertices then become phase-free, while the auxiliary Hamiltonian acquires the Berry term

$$\epsilon_O^{(m)} \rightarrow \epsilon_O^{(m)} - \zeta_\sigma \dot{\phi}(t). \quad (7)$$

Thus a gapped Hermitian auxiliary sector produces, in the adiabatic low-energy projection, a real spin-odd coherent shift. The passive non-Hermitian component appears only in the second step, when this shifted channel is embedded in the energy-dependent retarded Dirac bath. For the Gaussian projection the relevant adiabatic hierarchy is $\Omega \ll \epsilon_O^{(m)}$; the separate Kondo steady-state interpretation is restricted to the finite-drive window where the extracted screening scale exceeds the drive frequency. Because H_O is *bilinear* in $O_{k,m,\sigma}$, the modes $m \neq 0$ can be integrated out at the Gaussian level without approximation within the quadratic sector; the interaction U enters only through the slave-boson mean field on the impurity level (Sec. IV).

III. COARSE GRAINING AND EMERGENT $\mathcal{PT}$ SYMMETRY

This section isolates the bilinear coarse graining from the later slave-boson mean-field treatment of the local interaction. The raw retarded self-energy generated by the auxiliary angular modes is, in general, a complex retarded function; it is not assumed to be positive. The $\mathcal{PT}$ -active low-energy kernel is obtained by projecting the spin-dependent impurity self-energy onto a spin-conjugate pair z, z^* . This formulation keeps the microscopic origin of the passive relative gain-loss channel explicit while avoiding any assumption that the raw retarded envelope is real, positive, or sign definite.

A. Keldysh Path Integral and Angular-Mode Integration

We work within the Keldysh formalism [1], writing the partition function as a coherent-state path integral on the closed time contour. The bath and auxiliary angular modes enter quadratically and can be eliminated at the Gaussian level. The local Coulomb interaction on the impurity is not treated in this Gaussian step; it is incorporated later through the slave-boson mean-field approximation in Sec. V.

The Grassmann integration can be written explicitly. For fixed (k, θ, σ) , collect the auxiliary fields into a vector $O = (\dots, O_m, \dots)^T$ and write the quadratic contour action as

$$S_O + S_{cO} = \int_c dt dt' \bar{O}(t) A_O(t, t') O(t') + \int_c dt [\bar{c}(t) V_\sigma(t) O(t) + \bar{O}(t) V_\sigma^\dagger(t) c(t)] \quad (8)$$

The Gaussian identity for Grassmann variables gives

$$\int \mathcal{D}[\bar{O}, O] e^{i(S_O + S_{cO})} \propto \det A_O e^{iS_{\text{infl}}}, \quad (9)$$

$$S_{\text{infl}} = - \int_c dt dt' \bar{c}(t) \Xi_\sigma(t, t') c(t'), \quad (10)$$

$$\Xi_\sigma(t, t') = V_\sigma(t) g_O(t, t') V_\sigma^\dagger(t'). \quad (11)$$

Here $g_O = A_O^{-1}$ is the contour auxiliary propagator. This is the complete Gaussian step: because the auxiliary sector conserves particle number, there is no anomalous OO contraction. Keeping both vertices of Eq. (5) explicit, the normal auxiliary contraction gives

$$\Sigma_\sigma^{O,R}(t, t') = \sum_{m, m' \neq 0} v_{m\sigma}(t) g_{mm'}^{O,R}(t, t') v_{m'\sigma}^*(t'), \quad (12)$$

$$v_{m\sigma}(t) = V_m e^{i3m\theta} e^{i\zeta_\sigma \phi(t)}. \quad (13)$$

For the diagonal terms, and also for any Hermitian off-shell auxiliary matrix after summing conjugate paths, the phases enter as $e^{i\zeta_\sigma [\phi(t) - \phi(t')]}$. In Wigner coordinates this becomes $1 + O(\Omega/\epsilon_O)$ and gives a regular Hermitian principal-value shift. Equivalently, after the unitary rotation in Eq. (6), the same result is obtained from the spin-dependent level shift in Eq. (7).

To leading order in the slow drive and in the off-shell expansion, the auxiliary-dressed $m = 0$ bath therefore has the form

$$\Sigma_\sigma^{\text{bath},R}(\omega, t_{\text{av}}) = \Sigma_0^{\text{bath},R}(\omega) + \zeta_\sigma \Delta_{\text{aux}}(\omega, t_{\text{av}}) + O[(\Omega/\epsilon_O)^2]. \quad (14)$$

Here Δ_{aux} is real in the low-energy window. Microscopically one may write, schematically,

$$\Delta_{\text{aux}}(\omega, t_{\text{av}}) = - \dot{\phi}(t_{\text{av}}) \partial_{\epsilon_O} \sum_{m \neq 0} |V_m|^2 \times \mathcal{P} \frac{1}{\omega - \epsilon_O^{(m)}}, \quad (15)$$

up to form-factor and angular-overlap coefficients. If an auxiliary pole enters the conduction window, the corresponding delta-function part gives ordinary bath damping; it is not identified with the passive $\mathcal{PT}$ channel.

The bare impurity-bath hybridization W_k in Eq. (3) remains time independent. The drive reaches the impurity through the dressed bath propagator. In contour

notation the two Gaussian eliminations give

$$G_{c,\sigma}^{-1} = g_c^{-1} - \Xi_\sigma, \quad \Xi_\sigma = V_\sigma \circ g_O \circ V_\sigma^\dagger, \quad (16)$$

$$\Sigma_{d,\sigma} = W^\dagger \circ G_{c,\sigma} \circ W, \quad (17)$$

where $\circ$ denotes contour convolution. Thus the drive-dependent virtual path is $d \rightarrow c \rightarrow O \rightarrow c \rightarrow d$; no direct time dependence of W_k is required. The full Schur-complement derivation is given in the Supplemental Material.

In the frozen off-shell window, write $\Xi_\sigma^R = \Xi_0^R + \zeta_\sigma \delta \Xi^R$ with $\delta \Xi^R \in \mathbb{R}$. Expanding the dressed-bath resolvent gives the general first-order correction

$$\delta \Sigma_d^R = W^\dagger G_{c,0}^R \delta \Xi^R G_{c,0}^R W. \quad (18)$$

Although the auxiliary correction is real, the retarded continuum propagators are complex, so the transmitted impurity correction has both coherent and dissipative parts. Defining

$$\Sigma_{d,0}^R(\omega) = \Lambda(\omega) - i\bar{\Gamma}(\omega), \quad (19)$$

we write

$$\Sigma_{d,\sigma}^R = \Sigma_{d,0}^R + \zeta_\sigma \Delta_{\text{coh}} + i\zeta_\sigma \Gamma_{\text{PT}}. \quad (20)$$

For the nearly rigid scalar shift used in the analytic estimates, $\delta \Xi^R \simeq \Delta_{\text{aux}}$ and the spin-even auxiliary background varies slowly. Then

$$\Sigma_{d,\sigma}^R(\omega, t_{\text{av}}) \simeq \Sigma_{d,0}^R(\omega - \zeta_\sigma \Delta_{\text{aux}}) \simeq \Sigma_{d,0}^R - \zeta_\sigma \Delta_{\text{aux}} \partial_\omega \Sigma_{d,0}^R, \quad (21)$$

which gives

$$\Delta_{\text{coh}} = -\Delta_{\text{aux}} \partial_\omega \Lambda, \quad \Gamma_{\text{PT}} = \Delta_{\text{aux}} \partial_\omega \bar{\Gamma}. \quad (22)$$

Equation (18) is the general first-order result; Eq. (21) is its rigid-shift limit. All functions are evaluated in the active low-energy window. Equations (21)–(22) show why the Dirac bath is essential: for a flat wide band with $\partial_\omega \bar{\Gamma} = 0$, the same auxiliary shift would only produce a coherent spin splitting, not a passive $\mathcal{PT}$ decay imbalance. The SOC split bath also supplies the spin-momentum-locked helicity channels. In the chiral basis,

$$\bar{\Gamma}_\eta(\omega) = \pi \sum_k |W_{k\eta}|^2 \delta(\omega - \varepsilon_{c,\eta}(k)), \quad (23)$$

$$\varepsilon_{c,\eta}(k) = \alpha_b k^2 + \eta \lambda k. \quad (24)$$

so the SOC parameter λ controls the channel density of states, the slope $\partial_\omega \bar{\Gamma}_\eta$, and the form factor entering the projected EP diagnostics. Without SOC-induced helicity structure and energy-dependent Dirac spectral weight, the auxiliary shift remains a coherent real field rather than a passive relative decay channel.

B. Passive $\mathcal{PT}$ Core and Causal Detuning

The projected impurity self-energy is decomposed into a common bath part, a passive relative decay imbalance, and a Kramers–Kronig-related coherent detuning:

$$\Sigma_{\text{imp}}^R(\omega, t_{\text{av}}) = [\Lambda(\omega) - i\bar{\Gamma}(\omega)] \mathbb{I} + \Delta_{\text{coh}}(\omega, t_{\text{av}})\sigma_z + i\Gamma_{\text{PT}}(\omega, t_{\text{av}})\sigma_z. \quad (25)$$

The first line is a common Lamb shift and common decay. It is retained in the bath-controlled FDR, but it does not affect the relative EP structure. The second line contains two spin-odd terms with different symmetry status: $i\Gamma_{\text{PT}}\sigma_z$ is the passive relative decay imbalance, while $\Delta_{\text{coh}}\sigma_z$ is a real coherent detuning generated by the same causal retarded bath self-energy.

The SOC bath supplies the Hermitian spin-mixing channel. In the original spin basis, the local bath Green function has the matrix structure

$$g_b^R(k, \omega) = [\omega^+ - \epsilon_k - \lambda k \sigma_x]^{-1} = g_0(k, \omega)\mathbb{I} + g_x(k, \omega)\sigma_x, \quad (26)$$

with

$$g_x(k, \omega) = \frac{\lambda k}{(\omega^+ - \epsilon_k)^2 - \lambda^2 k^2}. \quad (27)$$

The principal-value part of this SOC off-diagonal bath dressing gives a real coherent spin-mixing coefficient,

$$\Delta_{\text{eff}}(\omega) = \sum_k |W_k|^2 \text{Re } g_x(k, \omega), \quad (28)$$

renormalized in the interacting problem by the slave-boson saddle point. The drive-induced shift cannot leak into this mixing channel at leading order: expanding the dressed bath propagator to first order in the spin-odd shift $\Delta_{\text{aux}}\sigma_z$ gives

$$g_b^R \sigma_z g_b^R = (g_0^2 - g_x^2) \sigma_z, \quad (29)$$

because $\{\sigma_x, \sigma_z\} = 0$, so the first-order correction is purely σ_z and Δ_{eff} remains a static (drive-independent) channel at this order, corresponding to the offset Δ_0 of Eq. (36); the isolated finite-drive EP of the generic projection is therefore the microscopically natural case. Thus, after subtracting the common damping, the general relative two-channel kernel is

$$h_{\text{rel}} = \Delta_{\text{eff}}\sigma_x + (\Delta_{\text{coh}} + i\Gamma_{\text{PT}})\sigma_z. \quad (30)$$

The passive- $\mathcal{PT}$ -symmetric core is obtained at the compensated detuning point $\Delta_{\text{coh}} = 0$:

$$h_{\mathcal{PT}} = \Delta_{\text{eff}}\sigma_x + i\Gamma_{\text{PT}}\sigma_z, \quad \sigma_x h_{\mathcal{PT}}^* \sigma_x = h_{\mathcal{PT}}. \quad (31)$$

The real term $\Delta_{\text{coh}}\sigma_z$ is not discarded. It is $\mathcal{PT}$ -odd under the same operation and therefore acts as a causal detuning floor. The pole splitting of Eq. (30) is

$$s^2 = \Delta_{\text{eff}}^2 + (\Delta_{\text{coh}} + i\Gamma_{\text{PT}})^2. \quad (32)$$

For real Δ_{eff} , Δ_{coh} , and Γ_{PT} , exact coalescence in the passive sector requires

$$\Delta_{\text{coh}}\Gamma_{\text{PT}} = 0, \quad \Delta_{\text{eff}}^2 + \Delta_{\text{coh}}^2 - \Gamma_{\text{PT}}^2 = 0. \quad (33)$$

Thus, for a nonzero relative decay imbalance, the exact passive EP occurs at the compensated point $\Delta_{\text{coh}} = 0$ and $|\Delta_{\text{eff}}| = |\Gamma_{\text{PT}}|$. Compensation is an operating-energy condition rather than a fine tuning: $\Delta_{\text{coh}} \propto \partial_\omega \Lambda$ changes sign and vanishes at a finite energy of order D_{uv}/e for the linear Dirac width, where $\partial_\omega \Gamma$ remains finite, while near the Fermi level the companion ratio is logarithmically enhanced, $|\Delta_{\text{coh}}/\Gamma_{\text{PT}}| \simeq (2/\pi) \ln(D_{\text{uv}}/|\omega|)$. Away from the compensated point, the Kramers–Kronig detuning prevents a literal pole coalescence: minimizing $|s|$ over Δ_{eff} at fixed Δ_{coh} and Γ_{PT} gives the causal floor

$$|s|_{\text{min}} = \sqrt{2|\Delta_{\text{coh}}|\Gamma_{\text{PT}}}, \quad (34)$$

with the same square-root non-analyticity as the interaction floor $s_{\text{eff}} = [U\beta_0\sqrt{F(\lambda)}]^{1/2}$ of the Supplemental Material, where $F(\lambda)$ is the SOC-overlap form factor: causality and interactions regularize the EP with the identical functional form. The figures should therefore be read as near-EP diagnostics of the co-decaying projected kernel, consistent with the bounded Z_η^{bio} and s_{min} regularization used in the BA-kernel analysis.

This two-step structure is the microscopic origin of the kernel used in the rest of the paper: off-shell auxiliary modes generate a real spin-odd shift; the energy-dependent Dirac bath converts that shift into a spin-dependent width; and the SOC bath provides the Hermitian spin-mixing channel. The numerical scans treat β_0 as a tunable frozen effective kernel control parameter. Microscopically,

$$\Gamma_{\text{PT}} \sim \Omega \frac{|V|^2}{\epsilon_0^2} \partial_\omega \bar{\Gamma}, \quad (35)$$

so the construction establishes the symmetry-allowed origin of the passive relative channel, while the plotted values explore the resulting frozen low-energy kernel.

IV. STEADY-STATE HAMILTONIAN AND EMERGENT NON-HERMITICITY

Given the passive spin-conjugate relative self-energy from Sec. III, this section derives the explicit 4×4 steady-state Hamiltonian $\tilde{H}(k)$ and identifies the structural origin of its exceptional points. A key step is the Hadamard rotation that diagonalizes the hybridization block, revealing that each chiral spinon mode couples exclusively to one bath channel—a property that later leads to the block-diagonal S -matrix used in the frozen-model Bethe-Ansatz diagnostic.

A. From Self-Energy to Steady-State Hamiltonian

After the Gaussian coarse-graining of the bath and auxiliary angular modes in Sec. III, and after introducing the slave-boson saddle point in Sec. V, the retarded spinon action takes the form $S_{\text{imp}}^R = \sum_k \tilde{f}[\omega\mathbb{I} - H_{\text{ss}}(k)]f$, where the steady-state Hamiltonian $H_{\text{ss}}(k)$ is defined as the inverse of the retarded Green's function at the saddle point: $H_{\text{ss}}(k) \equiv [G_{\text{ss}}^R(\omega, k)]^{-1}|_{\text{saddle}}$. The gauge-invariant slave-boson amplitude $r = |b_c|$ defines the natural low-energy scale $\tilde{\beta} = |\beta_0|r$ in the symmetric frozen-kernel convention used for the final numerical diagnostics. The sign of β_0 is then a drive or projection convention and does not change the physical low-energy amplitudes. The three channels that appear in the effective Hamiltonian have distinct microscopic origins: the anti-Hermitian passive relative gain-loss channel is the projected imaginary part of the retarded self-energy, the coherent splitting/flip is the projected Hermitian impurity channel, and the impurity-bath hybridization is the slave-boson-renormalized hybridization. We therefore use the channel-projected parametrization

$$\Gamma_{\text{PT}} = c_{\Gamma}\tilde{\beta}, \quad \Delta_{\text{eff}} = \Delta_0 + c_{\Delta}\tilde{\beta}, \quad V_{\text{eff}} = c_V\tilde{\beta}, \quad (36)$$

where Δ_0 is a possible static coherent splitting and c_{Γ} , c_{Δ} , and c_V are real low-energy projection coefficients. These coefficients should be understood as Wilson coefficients of the coarse-grained effective Hamiltonian, not as independent microscopic couplings inserted by hand. The anti-Hermitian spin-odd component of the retarded self-energy projects onto Γ_{PT} , the Hermitian spin-mixing component projects onto Δ_{eff} , and the remaining chiral-bath overlap determines V_{eff} . The common slave-boson scale is $\tilde{\beta} = |\beta_0||b_c|$ in the symmetric plotting convention, but $\mathcal{PT}$ symmetry only requires the projected coefficients to be real; it does not require $c_{\Gamma} = c_{\Delta} = c_V$. The minimal symmetric representative is $\Delta_0 = 0$ and $c_{\Gamma} = c_{\Delta} = c_V = 1$. A generic channel projection with $\Delta_0 \neq 0$ or unequal slopes separates an isolated finite-drive impurity EP from the artificial EP ray of the fully symmetric limit.

At the compensated passive- $\mathcal{PT}$ point $\Delta_{\text{coh}} = 0$, substituting the projected self-energy into the retarded Dyson equation and including the hybridization to the chiral bath channels gives, in the spin basis $(d_{\uparrow}, d_{\downarrow}, c_{k+}, c_{k-})$,

$$H_{\text{ss}}(k) = \begin{pmatrix} \tilde{\epsilon}_{\xi} + i\Gamma_{\text{PT}} & \Delta_{\text{eff}} & V_{\text{eff}}/\sqrt{2} & V_{\text{eff}}/\sqrt{2} \\ \Delta_{\text{eff}} & \tilde{\epsilon}_{\xi} - i\Gamma_{\text{PT}} & V_{\text{eff}}/\sqrt{2} & -V_{\text{eff}}/\sqrt{2} \\ V_{\text{eff}}/\sqrt{2} & V_{\text{eff}}/\sqrt{2} & \varepsilon_{c,+}(k) & 0 \\ V_{\text{eff}}/\sqrt{2} & -V_{\text{eff}}/\sqrt{2} & 0 & \varepsilon_{c,-}(k) \end{pmatrix}. \quad (37)$$

Here $\tilde{\epsilon}_{\xi} = \epsilon_{\xi} + \delta_c + \delta_{\Sigma}$ is the renormalized impurity level. The Lagrange multiplier δ_c enforces the slave-boson constraint, while δ_{Σ} is the real projected one-body shift from the principal-value self-energy in Sec. IIIB (and may be absorbed into ϵ_{ξ} by convention). The chiral bath dispersions are $\varepsilon_{c,\pm}(k) = \alpha_b k^2 \pm \lambda k$. The interaction U is not

an explicit two-body coupling in the 4×4 Gaussian kernel; it enters through the SBMF constraint and through the Schrieffer-Wolff / BA-kernel virtual charge energies used for the Kondo-scale estimate. For the finite- U Anderson interpretation, particle-hole symmetry is imposed by the common impurity-level choice $\epsilon_{\xi} = -U/2$, not by adding opposite $\pm U/2$ spin splittings. A real Zeeman field, when present, would enter as an independent spin splitting B_{imp} rather than as the Hubbard interaction.

The $\mathcal{PT}$ symmetry of $H_{\text{ss}}(k)$ is explicit for real Γ_{PT} , Δ_{eff} , and V_{eff} : under $P : \uparrow \leftrightarrow \downarrow$ combined with complex conjugation, $(\tilde{\epsilon}_{\xi} + i\Gamma_{\text{PT}}) \rightarrow (\tilde{\epsilon}_{\xi} - i\Gamma_{\text{PT}})^* = \tilde{\epsilon}_{\xi} + i\Gamma_{\text{PT}}$, while the real off-diagonal and hybridization entries are exchanged into themselves. Thus the $\mathcal{PT}$ structure is fixed by the projected self-energy channels and the gauge-invariant amplitude $r = |b_c|$, not by the phase of the slave-boson condensate.

B. Diagonal-Hybridization Rotation

The hybridization block of H_{ss} has Hadamard structure $(V_{\text{eff}}/\sqrt{2})\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$. The unitary rotation $U_2 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ acting on the impurity subspace defines chiral spinon modes $\tilde{d}_{\pm} = (d_{\uparrow} \pm d_{\downarrow})/\sqrt{2}$ and diagonalizes the hybridization block exactly:

$$U_2^{-1}h_{\text{imp}}U_2 = \begin{pmatrix} \tilde{\epsilon}_{\xi} + \Delta_{\text{eff}} & i\Gamma_{\text{PT}} \\ i\Gamma_{\text{PT}} & \tilde{\epsilon}_{\xi} - \Delta_{\text{eff}} \end{pmatrix}, \quad (38)$$

$$U_2^{-1}V_{dc} = V_{\text{eff}}\mathbb{I}, \quad (39)$$

where $h_{\text{imp}} = \begin{pmatrix} \tilde{\epsilon}_{\xi} + i\Gamma_{\text{PT}} & \Delta_{\text{eff}} \\ \Delta_{\text{eff}} & \tilde{\epsilon}_{\xi} - i\Gamma_{\text{PT}} \end{pmatrix}$ and $V_{dc} = (V_{\text{eff}}/\sqrt{2})\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$. The algebra is identical to the symmetric case: $\frac{V_{\text{eff}}}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}^2 = \frac{V_{\text{eff}}}{2} \cdot 2\mathbb{I} = V_{\text{eff}}\mathbb{I}$. The full rotated Hamiltonian in the $(\tilde{d}_{+}, \tilde{d}_{-}, c_{k+}, c_{k-})$ basis is:

$$\tilde{H}(k) = \begin{pmatrix} \tilde{\epsilon}_{\xi} + \Delta_{\text{eff}} & i\Gamma_{\text{PT}} & V_{\text{eff}} & 0 \\ i\Gamma_{\text{PT}} & \tilde{\epsilon}_{\xi} - \Delta_{\text{eff}} & 0 & V_{\text{eff}} \\ V_{\text{eff}} & 0 & \varepsilon_{c,+}(k) & 0 \\ 0 & V_{\text{eff}} & 0 & \varepsilon_{c,-}(k) \end{pmatrix}. \quad (40)$$

Three key properties of $\tilde{H}(k)$ follow:

- (i) *Diagonal hybridization*: $\tilde{d}_{+}$ hybridizes exclusively with the c_{k+} bath (amplitude V_{eff}) and $\tilde{d}_{-}$ with c_{k-} , independently. No cross-channel hybridization.
- (ii) *Impurity block*: the 2×2 frozen impurity block $h_d = \begin{bmatrix} \tilde{\epsilon}_{\xi} + \Delta_{\text{eff}} & i\Gamma_{\text{PT}} \\ i\Gamma_{\text{PT}} & \tilde{\epsilon}_{\xi} - \Delta_{\text{eff}} \end{bmatrix}$ contains a real coherent splitting Δ_{eff} and an anti-Hermitian relative gain-loss channel $i\Gamma_{\text{PT}}$. Both are measured in units of the gauge-invariant scale $\tilde{\beta} = |\beta_0||b_c|$, but their projection coefficients need not be identical.

(iii) *EP structure:* h_d has eigenvalues $E_{\pm} = \tilde{\epsilon}_{\xi} \pm s$ with $s = \sqrt{\Delta_{\text{eff}}^2 - \Gamma_{\text{PT}}^2}$. In the frozen diagnostic notation of Sec. IX, this is written as $s = \sqrt{\gamma^2 - \beta_0^2}$, with $\gamma \leftrightarrow \Delta_{\text{eff}}$ and $\beta_0 \leftrightarrow \Gamma_{\text{PT}}$ treated as independent coherent and relative gain-loss amplitudes. The local EP condition is $|\Gamma_{\text{PT}}| = |\Delta_{\text{eff}}|$; V_{eff} controls bath hybridization and bath-induced unfolding of this local condition.

a. Full 4×4 eigenvalue condition. The numerical eigenvalues and condition numbers in Figs. 1–3 are computed from the full channel-projected kernel of Eq. (40), not from the isolated impurity block alone. Eliminating the bath components gives the exact Schur-complement condition

$$\mathcal{P}(z, k) = \mathbf{p}_+(z, k)\mathbf{p}_-(z, k) + \Gamma_{\text{PT}}^2 = 0, \quad (41)$$

where

$$\mathbf{p}_{\eta}(z, k) = z - \tilde{\epsilon}_{\xi} - \eta\Delta_{\text{eff}} - \frac{V_{\text{eff}}^2}{z - \varepsilon_{c,\eta}(k)}, \quad \eta = \pm. \quad (42)$$

Equivalently, after clearing denominators, the full eigenvalues are roots of the quartic equation

$$\begin{aligned} & [\delta_{d,+}(z, k)\delta_{c,+}(z, k) - V_{\text{eff}}^2] [\delta_{d,-}(z, k)\delta_{c,-}(z, k) - V_{\text{eff}}^2] \\ & + \Gamma_{\text{PT}}^2 \delta_{c,+}(z, k)\delta_{c,-}(z, k) = 0, \end{aligned} \quad (43)$$

where

$$\delta_{d,\eta}(z, k) = z - \tilde{\epsilon}_{\xi} - \eta\Delta_{\text{eff}}, \quad \delta_{c,\eta}(z, k) = z - \varepsilon_{c,\eta}(k). \quad (44)$$

The full EP condition is therefore

$$\mathcal{P}(z_{\text{EP}}, k) = 0, \quad \partial_z \mathcal{P}(z_{\text{EP}}, k) = 0. \quad (45)$$

The simpler impurity-block expression in item (iii) is the transparent $V_{\text{eff}} \rightarrow 0$ limit and explains the local square-root mechanism; the plotted spectra use the full 4×4 matrix. In the opposite limit in which one chiral impurity-bath channel is isolated, the square root is

$$E_{\eta,\pm} = \frac{\epsilon_d + \epsilon_{k\eta}}{2} \pm \frac{1}{2} \sqrt{(\epsilon_d - \epsilon_{k\eta})^2 + 4V_{\eta}^R V_{\eta}^L}. \quad (46)$$

For a positive Hermitian product $V_{\eta}^R V_{\eta}^L > 0$ this gives the usual avoided crossing. If the product is negative or complex, the discriminant can instead vanish, producing a hybridization-induced non-Hermitian EP or crossover. In the revised gauge-fixed formulation, however, such a complex or sign-changing hybridization product is not identified with a Kondo scale unless it contributes to a positive screening eigenvalue of the biorthogonal exchange matrix.

b. Behaviour at $\beta_0 = 0$. In the minimal symmetric representative, $\Delta_0 = 0$, the exact origin has $\tilde{\beta} = 0$ and hence $\Gamma_{\text{PT}} = \Delta_{\text{eff}} = V_{\text{eff}} = 0$. The local impurity block then reduces to $\tilde{\epsilon}_{\xi} \mathbb{I}$. This is a normal twofold degeneracy, not a genuine exceptional point: the matrix is

diagonalizable and one may choose an orthonormal eigenbasis with $\kappa_{\text{imp}} = 1$. The apparent singularity of an EP-adapted normalization near this point is therefore a path-dependent diagnostic effect, not a physical finite-drive EP. In a generic projection with $\Delta_0 \neq 0$, the origin is instead a normal split impurity doublet, and an isolated local impurity EP occurs only when $|c_{\Gamma}\tilde{\beta}| = |\Delta_0 + c_{\Delta}\tilde{\beta}|$. Thus a finite offset or nonlinear channel projection converts the fully symmetric EP ray into an isolated finite-drive EP. Along a regular off-EP path the basis returns smoothly to a normal one and $\kappa_{\text{imp}} \rightarrow 1$.

The local interaction U does not by itself lift this trivial origin degeneracy in the spin-symmetric saddle point: it mainly renormalizes the scalar impurity level $\tilde{\epsilon}_{\xi}$ and the condensate amplitude. Only a spin-asymmetric term, such as an external Zeeman field or a spin-polarized real one-body shift proportional to $\langle n_{\downarrow} \rangle - \langle n_{\uparrow} \rangle$, would split the origin already at $\beta_0 = 0$. For this reason, the finite- $|\beta_0|$ condition-number peaks, where eigenvalue bifurcation and eigenvector coalescence occur away from the trivial normal degeneracy, are the physically relevant EP diagnostics in Figs. 1 and 2.

c. Interaction regularization of the EP. The interaction enters through the self-consistent SBMF parameters and shifts the frozen splitting s to an effective EP distance s_{eff} (Supplemental Material). Thus the condition number remains large but finite in the interacting calculation. We use $\kappa_{\text{imp}} = C/|s_{\text{eff}}|$ as a diagnostic of nonnormality near the EP rather than as an independent proof of a universal Kondo scale.

V. SLAVE-BOSON MEAN FIELD AND KELDYSH SELF-CONSISTENCY

The effective impurity problem originates from a correlated Anderson level with local repulsion U . We treat the constrained low-energy sector using large- N slave-boson mean-field theory [31, 34, 35] as a controlled saddle-point description of the Kondo regime ($T_b \ll T_K$). This section introduces the spinon operators $f_{\sigma}^{\dagger}$ that replace the physical $d_{\sigma}^{\dagger}$ after the slave-boson decoupling, derives the self-consistent equations for the condensate amplitude $r = |b_c|$ and Lagrange multiplier δ_c , and formulates the shifted resonance condition in gauge-invariant language.

A. Slave-Boson Representation

To handle the Coulomb repulsion U in the Keldysh framework, we use the large- N slave-boson (SBMF) representation [31, 34, 35]:

$$d_{\sigma}^{\dagger} = f_{\sigma}^{\dagger} b, \quad b^{\dagger} b + \sum_{\sigma} f_{\sigma}^{\dagger} f_{\sigma} = Q. \quad (47)$$

Here $f_{\sigma}^{\dagger}$ is the auxiliary spinon fermion carrying spin σ (the physical quasiparticle in the Kondo limit), and $b^{\dagger}$

is the slave boson enforcing no double occupancy. The parameter Q fixes the constraint normalization: $Q = 1$ gives the physical spin-1/2 infinite- U constraint, while $Q = Nq_{\text{occ}}$ denotes the large- N or channel-rescaled normalization. All saddle-point amplitudes quoted below are interpreted in the same normalization. The constraint is imposed via a Lagrange multiplier δ_c . At the saddle point $b \rightarrow b_c \in \mathbb{C}$, the phase of b_c is an internal $U(1)$ gauge choice. We therefore work in the gauge-fixed convention in which observables depend on the amplitude $r \equiv |b_c|$, and the spinon hybridization scale is rV_k .

The renormalized impurity level and gauge-invariant SBMF scale are then

$$\tilde{\epsilon}_\xi = \epsilon_\xi + \delta_c + \delta_\Sigma, \quad (48)$$

$$\tilde{\beta} = |\beta_0| |b_c| = |\beta_0| r \quad (49)$$

Here δ_Σ is the real principal-value one-body shift of the projected impurity level; if this shift is absorbed into the bare level, one may set $\delta_\Sigma = 0$ and reinterpret ϵ_ξ accordingly. The projected channels entering H_{ss} are subsequently defined by Eq. (36): $\Gamma_{\text{PT}} = c_\Gamma \tilde{\beta}$, $\Delta_{\text{eff}} = \Delta_0 + c_\Delta \tilde{\beta}$, and $V_{\text{eff}} = c_V \tilde{\beta}$. Thus $\tilde{\beta}$ fixes the natural slave-boson-renormalized scale, while Δ_0 and the projection coefficients encode, respectively, static coherent splitting, the anti-Hermitian self-energy channel, the drive-induced coherent splitting, and the impurity-bath form factor. We solve for b_c and δ_c self-consistently.

B. Saddle-Point Equations and Gauge-Invariant Amplitude

The Keldysh saddle-point conditions are obtained by varying the action with respect to b_c and δ_c on the quantum (Keldysh) branch (Supplemental Material). At the non-Hermitian fixed point, the equations take the form:

$$b_c \delta_c + 2i\beta_0 \sum_k [G_{\tilde{d}+,c+}^< + G_{\tilde{d}-,c-}^<] = 0, \quad (50)$$

$$r^2 - i \sum_{k,\pm} [G_{\text{ss}}^<(t,t)]_{\tilde{d}\pm,\tilde{d}\pm} = Q. \quad (51)$$

The value of Q is the same normalization as in Eq. (47); in particular, a rescaled large- N or two-channel numerical amplitude should not be identified directly with the bounded physical $Q = 1$ slave-boson amplitude. Here $G_{\tilde{d}\eta,c\eta}^<$ is the impurity-bath off-diagonal lesser Green's function and $[G_{\text{ss}}^<(t,t)]_{\tilde{d}\pm,\tilde{d}\pm}$ is the on-site occupation of the $\tilde{d}_\pm$ spinon. Although the slave-boson substitution $d_\sigma^\dagger = f_\sigma^\dagger b$ is introduced in the $\uparrow\downarrow$ basis, the saddle-point equations are equivalently written in the rotated $\tilde{d}_\pm = (d_\uparrow \pm d_\downarrow)/\sqrt{2}$ basis via the Hadamard rotation of Sec. IV B: at the saddle point $b \rightarrow b_c$ (a c-number), $\tilde{d}_\pm$ and the rotated spinons $\tilde{f}_\pm = (f_\uparrow \pm f_\downarrow)/\sqrt{2}$ are identical up to the overall factor b_c . The rewriting from the $\uparrow\downarrow$ basis to the $\pm$ basis is exact because (i) the Hadamard

rotation is unitary so $\sum_\sigma G_{f_\sigma c_\sigma}^< = G_{\tilde{f}_+,c_+}^< + G_{\tilde{f}_-,c_-}^<$, and (ii) the diagonal hybridization of $\tilde{H}(k)$ [Eq. (40)] kills all cross-channel terms $G_{\tilde{f}_+,c_-}^< = G_{\tilde{f}_-,c_+}^< = 0$ exactly. The full derivation in the $\uparrow\downarrow$ basis is given in Supplemental Material.

A central point, emphasized to avoid any gauge ambiguity, is that the physical amplitude $r = |b_c|$ is gauge-invariant, whereas the phase of b_c is an internal $U(1)$ slave-boson gauge degree of freedom ($b \rightarrow be^{i\theta}$, $f_\sigma \rightarrow f_\sigma e^{-i\theta}$). Consequently no observable conclusion is drawn from the phase of b_c . All spectra, condition numbers, and scale estimates below use the gauge-invariant amplitude $\tilde{\beta} = |\beta_0| r$ for the symmetric finite-drive diagnostics. If a numerical saddle is represented with a complex b_c , that phase is only a convention for the auxiliary fields and is not used as a physical source of the resonance shift.

C. Lesser Green's Function and Thermal Occupation

The physical lesser Green's function is fixed by the thermal bath. For a bilinear impurity problem coupled to a reservoir at temperature T_b , the Keldysh construction gives

$$\begin{aligned} G_{\text{bath}}^<(\omega) &= G^R(\omega) \Sigma_{\text{bath}}^<(\omega) G^A(\omega) \\ &= f(\omega) [G^R(\omega) - G^A(\omega)]. \end{aligned} \quad (52)$$

with $f(\omega) = [e^{\omega/T_b} + 1]^{-1}$. This bath-controlled object is the $G^<$ used for the physical occupations and for the SBMF self-consistency.

For comparison only, one can also assign Fermi occupations to the biorthogonal eigenmodes of $\tilde{H}(k)$,

$$G_{\text{eig}}^<(k) = iR(k) \text{diag}[f(\text{Re } \mathcal{E}_n)] L^\dagger(k), \quad (53)$$

where $L^\dagger \tilde{H} R = \text{diag}(\mathcal{E}_n)$ and $L^\dagger R = \mathbb{I}$. This eigenmode construction is not a physical thermal distribution function for the bath-coupled impurity; it is used only as an auxiliary diagnostic of nonorthogonality and must be interpreted with care near an EP.

D. Shifted Friedel-Type Resonance Condition

In the standard Hermitian slave-boson mean field at particle-hole symmetry ($\epsilon_\xi = -U/2$), the Kondo resonance is pinned to the Fermi level, $\omega_{\text{res}} = 0$, because the two spin occupations are equal and the saddle point fixes $\tilde{\epsilon}_\xi = 0$. In the present driven problem the coarse-grained self-energy produces spin-selective imaginary shifts $x_\sigma = \zeta_\sigma \Gamma_{\text{PT}}$ and therefore different effective widths in the two spin channels. The physical resonance position is then most conservatively described by the gauge-invariant condition

$$\omega_{\text{res}} \simeq \text{Re}(E_{\text{EP}}) \simeq \text{Re}(\tilde{\epsilon}_\xi), \quad (54)$$

valid when the low-energy line shape remains approximately Lorentzian. This is a shifted resonance condition rather than a claim that the gauge-dependent phase of b_c is observable. The shift is controlled by the renormalized level $\tilde{\epsilon}_\xi$, the spin-dependent widths, and the self-consistent amplitude $\tilde{\beta} = |\beta_0||b_c|$. The dashed black line in Fig. 3(d) marks this reference position. In the strongly non-Lorentzian regime the observed peak need not coincide exactly with this reference, and we therefore treat the deviation as a line-shape effect rather than as a violation of a sum rule. In the frozen kernel used for the plotted spectra, the local-moment reference has $\text{Re}(\tilde{\epsilon}_\xi) \simeq -1.0$ for $\epsilon_\xi = -U/2 = -1.0$; a fully self-consistent finite-temperature SBMF closure would replace this by its computed saddle-point value.

E. Near-EP SBMF Kondo-Scale Estimate

Near an isolated EP, the impurity-projected resolvent develops a large EP-active contribution associated with the coalescing left and right eigenvectors. We use the corresponding condition number κ_{imp} only as a diagnostic of this nonnormality and of the distance to the EP; it is not used as an independent definition of a Kondo temperature. The proper screening scale is controlled instead by the exchange eigenvalues of the channel-projected biorthogonal Schrieffer–Wolff matrix. We write

$$T_K^{\text{test}} = D_{\text{uv}} \exp \left[-\frac{1}{\chi_K j_{\text{scr}}} \right], \quad (55)$$

$$j_{\text{scr}} = \max \{ \text{Re } j_\alpha : j_\alpha \in \text{spec}(\rho_b J^{\text{bio}}), \text{Re } j_\alpha > 0 \}. \quad (56)$$

Here j_α are the eigenvalues of the dimensionless biorthogonal exchange matrix $\rho_b J^{\text{bio}}$, and only positive-real-part eigenvalues are retained as antiferromagnetic screening channels. The coefficient $\chi_K > 0$ fixes the conventional scaling normalization; all plotted estimates use $\chi_K = 1$. An EP-active sector can enhance the physical Kondo scale only if its residue feeds such a positive screening eigenvalue. By contrast, a hybridization induced EP generated by a negative scalar product $V^R V^L < 0$ can produce strong spectral nonnormality and a frozen non-Hermitian crossover scale, but it should not automatically be identified with an enhanced thermodynamic Kondo temperature. Supplemental Material gives the corresponding frozen-model scale estimate and explains why the spin-selective sign $x_\uparrow = -x_\downarrow$ alone does not split the scale unless the channel widths, real detunings, or exchange eigenvalues are unequal.

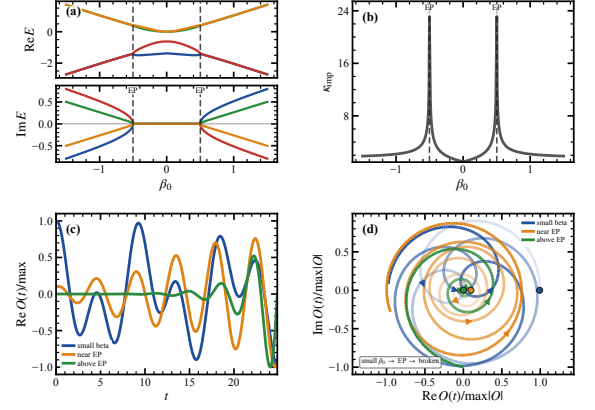


FIG. 1. **Full 4×4 non-Hermitian impurity kernel: spectrum, condition number, and real-time dynamics.** All panels use the gauge-fixed frozen-kernel convention $\tilde{\beta} = |\beta_0|r$ with $\epsilon_\xi = -U/2 = -1.0$, $U = 2.0$, and $T_b = 0.1$. The plotted eigenvalues and eigenvectors are computed from the full rotated 4×4 kernel of Eq. (40), not from the isolated impurity block. **(a)** Real and imaginary parts of the four eigenvalues of $\tilde{H}(k=0)$ versus β_0 . The representative finite-drive local EPs occur at $|\beta_0| \simeq 0.50$; double-root tracking gives $\beta_{0,\text{proj}}^{\text{EP}} = 0.4975$, while the discrete grid peaks at the adjacent point $\beta_0 = 0.5025$ ($\Delta\beta_0 = 0.0075$). **(b)** Condition number $\kappa_{\text{imp}} = \|R\| \|R^{-1}\|$, using the same eigenvector normalization throughout the scan. The finite- $|\beta_0|$ peaks locate eigenvector coalescence; the origin is a normal degeneracy and is not interpreted as an EP. **(c)** Survival-amplitude dynamics for representative points below, at, and above the EP, plotted on a common solver time window. **(d)** Corresponding phase portraits. The change from oscillatory motion to critical coalescence and then to broken-regime overdamped response is the time-domain signature of the 4×4 finite-drive EP.

VI. SPECTRAL PROPERTIES AND PHYSICAL REGIMES

Unless stated otherwise, the final numerical figures use the uniform local-moment convention

$$\epsilon_\xi = -U/2 = -1.0, \quad U = 2.0, \quad T_b = 0.1, \quad (57)$$

with the gauge-fixed projected scale $\tilde{\beta} = |\beta_0|r$. The projected double-root value is $\beta_{0,\text{proj}}^{\text{EP}} = 0.4975$; the discrete condition-number grid peaks at the adjacent point $\beta_0 = 0.5025$. Figure 1 displays the corresponding pair at $|\beta_0| \simeq 0.50$.

Figures 1 and 2 display the dynamical fingerprints of the EP-active projected impurity block and of the relative gain–loss-only diagnostic limit, respectively. In the projected Kondo-regime kernel ($\epsilon_\xi = -1.0 = -U/2$, $U = 2.0$, $T_b/T_K < 1$), the representative coefficients place a symmetric local EP pair at $|\beta_0| \simeq 0.5$, accompanied by a condition-number peak and by a crossover from oscillatory to overdamped local dynamics. Removing the coherent splitting channel ($\gamma = 0$, Fig. 2) removes the coherent local impurity EP mechanism. The remaining passive gain–loss dynamics is retained only

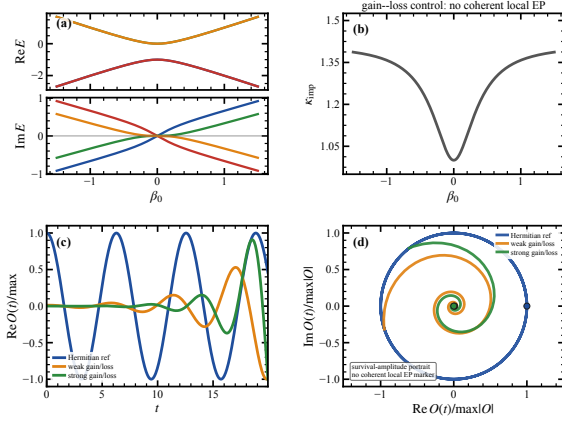


FIG. 2. **Full 4×4 relative gain-loss-only control.** The same impurity-level convention is used as in Fig. 1, $\epsilon_\xi = -U/2 = -1.0$ with $U = 2.0$ and $T_b = 0.1$, but the coherent local splitting/flip channel is switched off. This isolates passive relative gain-loss and bath dressing from the coherent impurity-EP mechanism. (a) Real and imaginary parts of the four eigenvalues of the frozen 4×4 kernel. (b) Condition-number diagnostic. Any residual structure is a passive non-Hermitian bath-dressing diagnostic and is not interpreted as a coherent finite-drive local impurity EP. (c) Survival-amplitude dynamics for the Hermitian reference, weak relative gain-loss, and strong relative gain-loss cases, plotted on a common time window. (d) Phase portraits for the same three cases. These trajectories are shown in a normalized or co-decaying representation; they should not be read as conservation of the Euclidean norm of the isolated relative gain-loss block.

as a diagnostic contrast: it should not be interpreted as a separate positive-screening Kondo scale enhancement.

a. Condition-number convention. All condition numbers shown below are computed from the Euclidean right-eigenvector matrix of the full 4×4 channel-projected kernel at the plotted k slice, using a single normalization convention for all values of β_0 . We use κ_{imp} to locate finite- $|\beta_0|$ eigenvector coalescence and to track the distance to the EP. Its absolute magnitude is not used as a standalone physical observable or as an independent definition of the Kondo scale; the physical diagnostics are the spectral line shapes, low-energy scale estimates, occupied fluctuation-dissipation-relation (FDR)-weighted spectra, and frozen channel-projected amplitudes extracted with the same gauge-invariant convention $\hat{\beta} = |\beta_0|/|b_c|$.

A. Phase-Selective EP Enhancement

Figure 3 should be read as a compact frozen channel-projected spectral diagnostic, not as a full finite-temperature SBMF phase diagram. The condition-number and T_b/T_K regime classifiers are discussed in the text and caption rather than assigned separate panels. Panel (a) deliberately shows only the non-circular

SBMF positive spectral-weight-transfer diagnostic on its own normalized scale. This avoids forcing a window-integrated SBMF spectral measure and the analytic SW/BA screening estimates to share an artificial absolute normalization. The full relative comparison of SW, BA, and the SBMF transfer ratio is given in Supplemental Fig. S3. The exported data also include the peak-resolved HWHM and re-solved FDR-window temperature diagnostics, but these are not used as main scales because they can become ill-defined or non-monotonic when several peaks overlap.

In the local-moment reference, particle-hole symmetry is imposed uniformly by $\epsilon_\xi = -U/2 = -1.0$ with $U = 2.0$. The Hubbard interaction is not inserted as opposite $\pm U/2$ spin splittings in the 4×4 quasiparticle kernel. The impurity-like spectral feature is therefore referenced to $\omega \simeq \text{Re}(\tilde{\epsilon}_\xi) = -1.0000$ in the frozen closure. Moving ϵ_ξ close to zero would move the coalescence toward the Fermi level, but that would be a resonant or mixed-valence reference rather than the Kondo local-moment convention used for the final figures.

VII. FLUCTUATION-DISSIPATION RELATION AND SPECTRAL FUNCTIONS

For a non-Hermitian system, two inequivalent constructions of $G^<$ arise and it is essential to distinguish them. Figure 4 separates these objects explicitly. The impurity and bath FDR ratios constructed from the reservoir self-energy follow the same thermal Fermi function on spectral support. The eigenmode-based construction is displayed in a separate diagnostic panel because it is not a bath-imposed distribution function; deviations of this diagnostic are therefore not a physical violation of the fluctuation-dissipation relation.

a. Bath-based construction (physical). The physical lesser Green's function is fixed by the thermal bath self-energy. For a bilinear impurity-bath coupling,

$$G_{\text{phys}}^<(\omega) = G^R(\omega) \Sigma_{\text{bath}}^<(\omega) G^A(\omega). \quad (58)$$

If the bath is thermal, then

$$\Sigma_{\text{bath}}^<(\omega) = f(\omega) [\Sigma_{\text{bath}}^R(\omega) - \Sigma_{\text{bath}}^A(\omega)]. \quad (59)$$

Combining this with the Dyson identity for the bath-induced broadening gives

$$G_{\text{phys}}^<(\omega) = f(\omega) [G^R(\omega) - G^A(\omega)]. \quad (60)$$

Thus the physical impurity FDR is inherited from the reservoir and is the $G^<$ used in the SBMF self-consistency.

b. Eigenmode-based construction (comparison). $G_{\text{eig}}^< = iR \text{diag}(f(\text{Re } \mathcal{E}_n)) L^\dagger$ assigns Fermi occupations $f(\text{Re } \mathcal{E}_n)$ to the non-Hermitian eigenvalues $\mathcal{E}_n$. Because the eigenvalues are complex, this construction does not correspond to the thermal occupations imposed by the

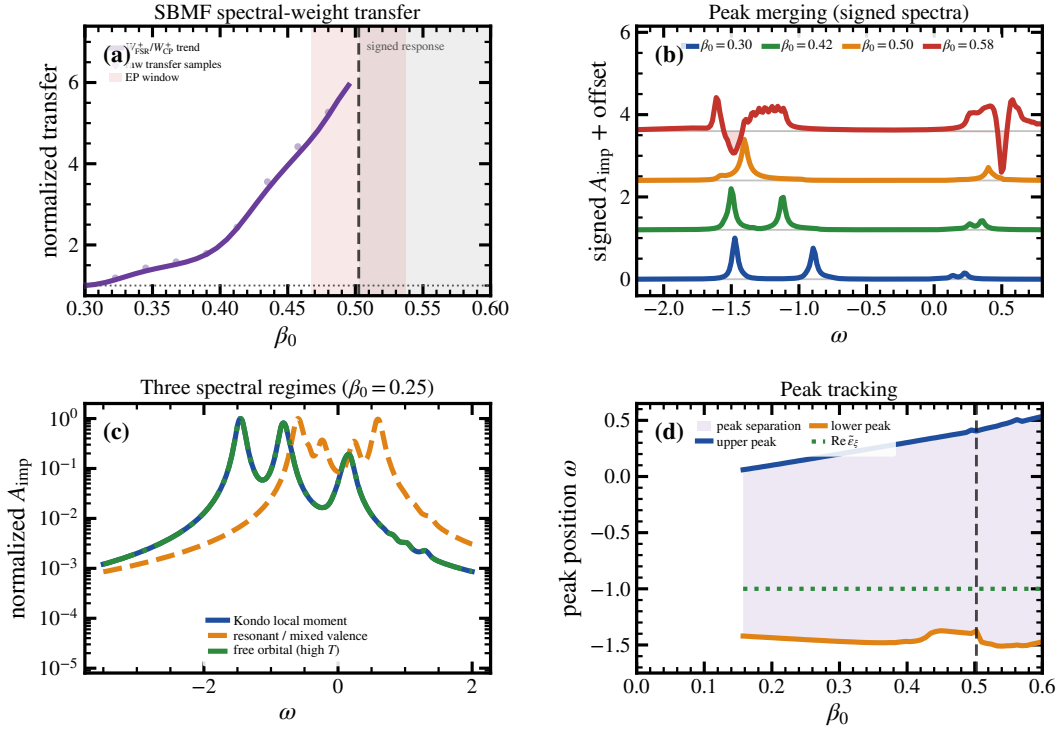


FIG. 3. **Compact frozen regime diagnostics: scales, peak merging, three-regime spectra, and peak tracking.** All panels use $\epsilon_\xi = -U/2 = -1.0$, $U = 2.0$, and the gauge-fixed scale $\tilde{\beta} = |\beta_0|r$. The positive-branch EP marker is $\beta_{0,\text{proj}}^{\text{EP}} = 0.4975$ in the plotted scan. **(a)** Independent SBF spectral-weight-transfer check. The plotted quantity is the positive spectral-weight transfer ratio $\mathcal{W}_{\text{FSR}}^+/\mathcal{W}_{\text{CP}}^+$, normalized to the stable low-drive reference $\beta_{0,\text{ref}} = 0.30$. Here $\mathcal{W}_{\text{FSR}}^+$ is the positive spectral weight in a window around the shifted FSR/Kondo feature $\omega_{\text{res}} \simeq \text{Re}\tilde{\epsilon}_\xi$, while $\mathcal{W}_{\text{CP}}^+$ is the corresponding positive weight in the central-peak window near $\omega = 0$. The faint points are raw window-integrated values and the solid line is a local trend. The ratio is meaningful only once the central-peak window carries non-negligible positive weight, so the plotted range starts at the stable low-drive plateau on which the reference is taken; at smaller β_0 the central peak has not yet formed and the raw ratio is dominated by the vanishing denominator (the full range is included in the exported data). The diagnostic terminates at the negative-signed-weight onset $\beta_0 \simeq 0.50$, i.e. at the EP, beyond which the signed response is reported separately. This construction avoids assigning a linewidth when several peaks overlap or the signed projected response becomes non-Lorentzian. The analytic SW/BA screening estimates are not overlaid here; they are compared with the same SBF transfer diagnostic on a common relative scale in the Supplemental Material. **(b)** Peak-merging diagnostics extracted from the same frozen spectra: the centroid separation, the upper- and lower-window widths, and the local EP-distance $|s_{\text{eff}}|$ are normalized to their panel maxima so the coalescence/redistribution trend is visible even when the raw peaks are close. **(c)** Broadened normalized retarded spectra for the screened local-moment, resonant/mixed-valence, and high-temperature free-orbital references. When the same frozen retarded kernel is used, regime distinctions come from T_b/T_K and from the occupied FDR-weighted spectrum rather than from $A(\omega)$ alone. **(d)** Peak positions versus β_0 . The reference line is $\text{Re}(\tilde{\epsilon}_\xi) = -1.0000$. In the scan, the upper-window centroid ranges from -1.60×10^{-2} to 1.63×10^{-1} , while the lower-window centroid ranges from -1.181 to -0.709 ; the apparent peak separation is largest near the EP because the signed frozen spectral response is strongly non-Lorentzian there.

bath. Consequently $G_{\text{eig}}^<$ is not interpreted as a physical distribution function. It is retained only as an auxiliary diagnostic of the nonorthogonal eigenbasis and is not used in the SBF self-consistency or in the physical FDR analysis.

c. Impurity spectral function. The impurity spectral function is evaluated from the retarded and advanced Green's functions as

$$A_{\text{imp}}(\omega) = -\frac{1}{2\pi} \text{Im Tr}[G^R(\omega) - G^A(\omega)]_{\text{imp}}. \quad (61)$$

We use this bath-consistent construction throughout the spectral analysis. In contrast, using only $-\text{Im Tr } G^R/\pi$

can introduce spurious features from relative-kernel poles in a non-Hermitian representation. The spectral results below are therefore computed with the same retarded, advanced, and bath-controlled lesser functions used in the self-consistency.

d. Gauge convention for spectral functions. The physical self-energy from the coarse-graining is first decomposed into a common bath shift/damping, the passive $\mathcal{PT}$ core of Eq. (31), and the Kramers–Kronig detuning of Eq. (30). The off-shell auxiliary sector supplies the real spin-odd shift, while the energy-dependent Dirac bath converts this shift into the relative spin-dependent width Γ_{PT} . The effective Hamiltonian is therefore writ-

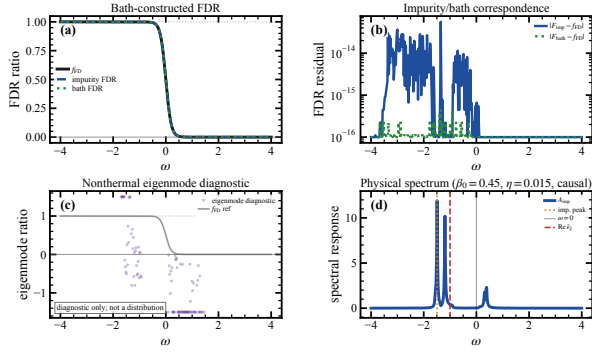


FIG. 4. **Bath-controlled FDR, eigenmode diagnostic, and physical spectral response.** The uniform impurity convention is $\epsilon_\xi = -U/2 = -1.0$, $U = 2.0$, and $T_b = 0.1$. The bath-based FDR construction is the physical one used for occupations and SBMF self-consistency. (a) Impurity and bath FDR ratios obtained from $G_{\text{phys}}^< = f_{\text{FD}}(\omega)[G^R - G^A]$ on spectral support. Both follow $f_{\text{FD}}(\omega)$; in the exported diagnostic file the maximum residual of the bath-constructed ratio from f_{FD} is below 10^{-12} . (b) Absolute FDR residual for the bath-constructed physical ratio; the exported diagnostic gives a maximum residual below 10^{-12} . (c) Eigenmode-based ratio obtained by assigning occupations to non-Hermitian eigenmodes. This object is nonthermal and is shown only as a non-normality/eigenbasis diagnostic; it is not used in the saddle-point equations or in the physical FDR analysis. (d) Physical impurity spectral response with the Fermi level $\omega = 0$, the frozen impurity reference $\text{Re}(\tilde{\epsilon}_\xi) = -1.0000$, and the observed spectral peak marker. When the frozen kernel is instead evaluated on the PT-broken side, a negative signed response should be reported explicitly as a causality diagnostic, not as a positive physical DOS.

ten in terms of the real, gauge-invariant amplitude $\tilde{\beta} = |\beta_0||b_c|$ and the projected channels $\Gamma_{\text{PT}} = c_\Gamma\tilde{\beta}$, $\Delta_{\text{eff}} = \Delta_0 + c_\Delta\tilde{\beta}$, and $V_{\text{eff}} = c_V\tilde{\beta}$. The phase of the auxiliary slave-boson field is not used separately in spectral calculations. With this convention, the same H_{ss} is used for eigenvalues, condition numbers, and spectral functions, and no observable is attributed to the gauge-dependent phase of b_c .

VIII. REAL-TIME DYNAMICS

The real-time diagnostic is the complex survival amplitude

$$O(t) = \langle d_\uparrow | e^{-iHt} | d_\uparrow \rangle, \quad (62)$$

evaluated for the same gauge-fixed frozen kernel used in the spectral plots. We show both $\text{Re } O(t)$ and the phase portrait $[\text{Re } O(t), \text{Im } O(t)]$. This is a non-unitary propagator diagnostic of the local EP; it is not used as a normalized thermal expectation value. For Fig. 1(c),(d) we use $\epsilon_\xi = -U/2 = -1.0$, $U = 2.0$, $T_b = 0.1$, and $\beta_{0,\text{proj}}^{\text{EP}} \approx 0.5$. The evolution shows a direct dynamical signature of the EP:

- *Below EP* ($\beta_0 < \beta_{0,\text{proj}}^{\text{EP}}$, local splitting real): eigenvalues are non-degenerate; the evolution shows underdamped oscillations with a frequency set by $\text{Re}(E_+ - E_-) = 2\text{Re } s$ and a decay rate set by $\text{Im}(E_\pm)$.
- *Above EP* ($\beta_0 > \beta_{0,\text{proj}}^{\text{EP}}$, local splitting imaginary): eigenvalues have opposite imaginary parts; the gain mode dominates and the evolution shows overdamped decay.
- *Without flip* ($\gamma = 0$, Fig. 2): the isolated relative gain-loss block has eigenvalues $\tilde{\epsilon}_\xi \pm i\Gamma_{\text{PT}}$ and is therefore in the broken sector for $\Gamma_{\text{PT}} \neq 0$. The plotted bounded trajectories are normalized or co-decaying diagnostics of the full bath-coupled kernel, not evidence for conservation of the Euclidean norm.

The contrast between the normalized no-flip diagnostic and the spiraling dynamics with coherent splitting demonstrates that both projected impurity channels of $\tilde{H}$ —the coherent splitting Δ_{eff} and the relative decay channel $i\Gamma_{\text{PT}}$ —are necessary for the full finite-drive EP structure. In the channel-projected parametrization, these channels are measured relative to the common scale $\tilde{\beta}$ but should be viewed as distinct projected self-energy components. Equality of their coefficients is a symmetric representative limit, not a symmetry requirement.

IX. BIORTHOGONAL BETHE ANSATZ

The saddle-point treatment of Sec. V motivates the gauge-invariant scale $\tilde{\beta} = |\beta_0||b_c|$ and the renormalized level $\tilde{\epsilon}_\xi$. The Bethe-Ansatz analysis below is a frozen effective-model diagnostic: γ denotes the coherent impurity splitting/flip channel, while β_0 denotes the projected relative gain-loss channel in the minimal frozen notation. The general projected description is $\Gamma_{\text{PT}} = c_\Gamma\tilde{\beta}$, $\Delta_{\text{eff}} = \Delta_0 + c_\Delta\tilde{\beta}$, and $V_{\text{eff}} = c_V\tilde{\beta}$. The local EP is controlled by the competition between Γ_{PT} and Δ_{eff} , while V_{eff} controls bath coupling and the positive screening hybridization. The diagonal hybridization structure [property (i) of Sec. IV B] is the key that makes the frozen- $\tilde{\beta}$ problem analytically tractable: it forces the two-body S -matrix to be block-diagonal ($S^{LR} = S^{RL} = 0$), providing the channel separation used in the Bethe-Ansatz construction of the frozen effective model. For the frozen diagnostic phase diagram, β_0 (gain/loss amplitude) and γ (coherent flip amplitude) are kept as independent projection channels. Their equality is a special symmetric representative, not a general consequence of the slave-boson saddle. The frozen-BA approximation is used only as a low-energy analytic screening diagnostic, most directly when $T_b \ll T_K$ and hybridization fluctuations about the saddle point are small.

A. Block-Diagonal S -Matrix from Diagonal Hybridization

The diagonal hybridization [property (i) of Sec. IV B] is the key that makes the frozen effective model channel-separated. Because $\tilde{d}_+$ couples only to c_{k+} and $\tilde{d}_-$ only to c_{k-} , the two-body scattering is channel-diagonal: bath + electrons can only scatter from the $\tilde{d}_+$ impurity state and vice versa. The internal relative channel $i\beta_0$ is an excitation that mixes the two impurity states, but it does not connect incoming bath + electrons to outgoing bath – electrons. This follows because the hybridization vertices preserve the bath channel index, so impurity mixing does not induce cross-channel propagation of bath electrons. This is verified by the two-body contact algebra (Supplemental Material): the off-diagonal scattering amplitudes e_{+-} and e_{-+} satisfy homogeneous equations [Eqs. (G2) and (G4)] with no external bath source terms. Together with the density-density interaction $H_U = U n_{d+} n_{d-}$ (which preserves channel labels), this gives the block-diagonal S -matrix:

$$S^{LR} = S^{RL} = 0, \quad S_{\text{tot}} = \begin{pmatrix} S^{RR} & 0 \\ 0 & S^{LL} \end{pmatrix}. \quad (63)$$

The biorthogonal impurity problem therefore separates into independent right and left sectors, which is the structural input used for the Bethe-Ansatz construction below.

B. Renormalized Contact Denominators and Breit–Wigner S -Matrix

The two-body contact algebra (Supplemental Material) gives the renormalized denominators through the relative channel β_0 :

$$\mathbf{a} = \tilde{\epsilon}_\xi + \gamma - E, \quad \mathbf{b} = \tilde{\epsilon}_\xi - \gamma - E, \quad (64)$$

$$\tilde{\mathbf{a}} = \mathbf{a} + \frac{\beta_0^2}{\mathbf{b}}, \quad \tilde{\mathbf{b}} = \mathbf{b} + \frac{\beta_0^2}{\mathbf{a}}. \quad (65)$$

Here $\mathbf{a}$ and $\mathbf{b}$ are energy denominators for the $\tilde{d}_+$ and $\tilde{d}_-$ impurity levels respectively; the bath dispersion $\epsilon_{c,\pm}(k)$ enters the contact equations through the bath wavefunctions $\Phi_\pm$, not through $\mathbf{a}$ or $\mathbf{b}$. Each diagonal block of the S -matrix takes the Breit–Wigner form:

$$S_{\eta\eta}^{XX}(k) = \frac{k - \tilde{\epsilon}_{d,\eta} - i\tilde{\Gamma}_\eta}{k - \tilde{\epsilon}_{d,\eta} + i\tilde{\Gamma}_\eta}, \quad X \in \{R, L\}, \quad \eta \in \{+, -\}, \quad (66)$$

where $\tilde{\epsilon}_{d,\eta} = \epsilon_{d,\eta} + \Lambda_\eta^{\text{PV}}$ is the bath-shifted impurity level with one-body shift $\Lambda_\eta^{\text{PV}} = (V_\eta)^2 \sum_k (\epsilon_{d,\eta} - \epsilon_{c,\eta}(k))^{-1}$. For reciprocal Hermitian hybridization the resonance width is $\tilde{\Gamma}_\eta = \pi \rho_{b,\eta}(0) |V_\eta|^2$. In a genuinely non-Hermitian or nonreciprocal channel this is replaced by the biorthogonal product

$$\tilde{\Gamma}_\eta^{\text{bio}} = \pi \rho_{b,\eta}(0) V_\eta^R V_\eta^L. \quad (67)$$

The density $\rho_{b,\eta}(0)$ used here is the local radial-channel density of states of the frozen channel-projected effective model, evaluated at the active resonance after the chiral rotation and low-energy linearization. It is not the bare two-dimensional Dirac point density $\rho_{\text{bath}}(\omega) \sim |\omega|/\lambda^2$, which vanishes at the undoped Dirac point. In the frozen BA diagnostic we work in a finite scattering window where $\rho_{b,\eta}$ is slowly varying and may be replaced by the constant $\rho_{b,\eta}(0) \simeq 1/k_{\text{max}}$. The full numerical spectral calculations retain the bath dispersion explicitly.

C. Integrability via Yang–Baxter Equation

Because the diagonal hybridization [Eq. (40)] decouples the + and – channels completely, the S -matrix in each diagonal block is a *scalar* Breit–Wigner [Eq. (66)]. Scalar S -matrices trivially satisfy the Yang–Baxter equation: for c-number $S_{\eta\eta}^X(k)$, the three-body relation

$$\begin{aligned} & S_{\eta\eta}^X(k_1 - k_3) S_{\eta\eta}^X(k_2 - k_3) S_{\eta\eta}^X(k_1 - k_2) \\ &= S_{\eta\eta}^X(k_1 - k_2) S_{\eta\eta}^X(k_2 - k_3) S_{\eta\eta}^X(k_1 - k_3) \end{aligned} \quad (68)$$

holds identically by commutativity of scalar multiplication, establishing channel-separated scattering in the frozen model.

Supplemental Material gives the same statement in the contact-projector notation. Since each frozen block is a scalar Breit–Wigner amplitude, YBE factorization follows from channel separation and scalar commutativity alone. No additional Baxterization functional equation is required. Bath dispersions enter only as diagonal one-body phases in the Bethe equations and do not spoil factorization.

D. Bethe Ansatz Equations

The charge rapidities $\{k_j^R\}$ in the right sector satisfy:

$$e^{ik_j^R \mathcal{L}} = \prod_{\ell \neq j} S^{RR}(k_j^R - k_\ell^R) \prod_{\eta,a} \frac{\mu_a^{R(\eta)} - k_j^R + \tilde{\epsilon}_{d\eta} + i\tilde{\Gamma}_\eta^R/2}{\mu_a^{R(\eta)} - k_j^R + \tilde{\epsilon}_{d\eta} - i\tilde{\Gamma}_\eta^R/2}, \quad (69)$$

and the auxiliary rapidities $\{\mu_a^{R(\eta)}\}$ satisfy:

$$\begin{aligned} & \prod_j \frac{\mu_a^{R(\eta)} - k_j + \tilde{\epsilon}_{d\eta} + i\tilde{\Gamma}_\eta^R/2}{\mu_a^{R(\eta)} - k_j + \tilde{\epsilon}_{d\eta} - i\tilde{\Gamma}_\eta^R/2} \\ &= \prod_{b \neq a} \frac{\mu_a^{R(\eta)} - \mu_b^{R(\eta)} + i\tilde{\Gamma}_\eta^R}{\mu_a^{R(\eta)} - \mu_b^{R(\eta)} - i\tilde{\Gamma}_\eta^R}. \end{aligned} \quad (70)$$

Left-sector equations follow by $R \rightarrow L$. Written with symmetric $\pm i\tilde{\Gamma}_\eta/2$ shifts, these equations reduce to the standard Hermitian Anderson impurity quantization conditions when $V_\eta^R = V_\eta^L$ and the relative non-Hermitian channel is set to zero. The bare $U = 0$ EP manifests

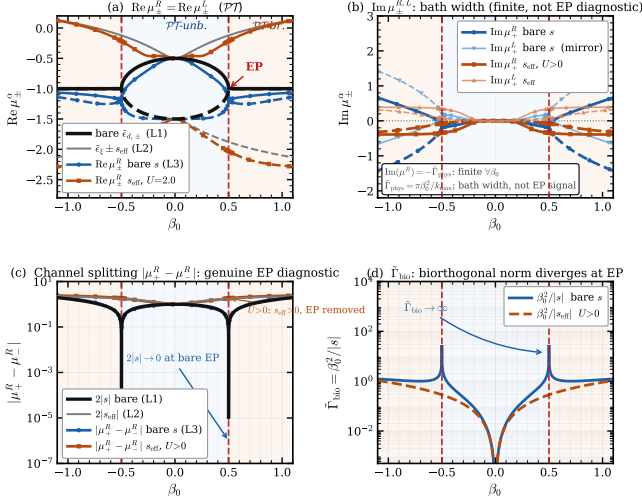


FIG. 5. Biorthogonal BA rapidities and EP structure. Three layers per panel represent distinct levels of physics ($\epsilon_{\xi} = -U/2 = -1.0$, $U = 2.0$, $\gamma = \beta_0 = 0.5$ at bare EP, $\lambda = 0.50$, $k_{\text{max}} = \pi/4$, D-SOC² form factor $F = 1 - (\lambda/k_{\text{max}})^2 \approx 0.595$). Vertical red dashed lines mark the bare EP at $|\beta_0| = \gamma = 0.5$. *Layer 1 (thick black):* bare h_d eigenvalues $E_{\pm}^{\text{bare}} = \tilde{\epsilon}_{\xi} \pm s$, $s = \sqrt{\gamma^2 - \beta_0^2}$. This is the only layer with a genuine EP (at $\beta_0 = \gamma$, where $s = 0$). *Layer 2 (grey):* U -corrected levels $\tilde{\epsilon}_{\xi} \pm s_{\text{eff}}$ from the quartic (Supplemental Material); $s_{\text{eff}} > 0$ for all β_0 when $U > 0$ and $F(\lambda) > 0$ (EP avoided). *Layer 3 (blue circles: bare s ; orange squares: s_{eff}):* Bethe-dressed rapidities $\mu_{\pm}^{R,U} = (\tilde{\epsilon}_{\xi} \pm s_{\text{eff}}) + \Lambda_{\pm}^{\text{PV}}$ using the physical S-matrix resonance width $\tilde{\Gamma}_{\eta} = \pi V_{\eta}^2 \rho_{b,\eta}(0) = \pi \beta_0^2/k_{\text{max}}$ (for this frozen diagnostic $r = 1$, so $V_{\eta} = \tilde{\beta} = |\beta_0|$) in the bath shift $\Lambda_{\pm}^{\text{PV}}$; $\mu^L = (\mu^R)^*$ by $\mathcal{PT}$ symmetry. **(a)** Real parts vs β_0 : $\text{Re } \mu^R = \text{Re } \mu^L$ by $\mathcal{PT}$ symmetry; only the bare $U = 0$ layer coalesces at the bare EP, while the $U > 0$ layers remain split. **(b)** Imaginary parts vs β_0 : $\text{Im } \mu^R = -\tilde{\Gamma}_{\text{phys}} = -\pi \beta_0^2/k_{\text{max}}$ is the physical bath decay width, finite everywhere for all β_0 . $\text{Im } \mu^L = -\text{Im } \mu^R$ by $\mathcal{PT}$ (mirror symmetry). The imaginary part is *not* an EP diagnostic; see panel (c). **(c)** Channel splitting $|\mu^{R(+)} - \mu^{R(-)}|$ (log scale): the genuine EP diagnostic (text after Eq. 70). Three reference layers are shown: $2|s|$ (Layer 1, black), $2|s_{\text{eff}}|$ (Layer 2, grey), and the bath-dressed splitting (Layer 3, coloured). Bare s touches zero at $\beta_0 = \gamma$; U -corrected s_{eff} stays finite (EP regularised by interactions). **(d)** Biorthogonal width $\tilde{\Gamma}_{\text{bio}} = \beta_0^2/s$ vs β_0 (log scale): diverges at the bare EP ($s \rightarrow 0$), driving the biorthogonal norm divergence and BA-kernel nonnormality enhancement; regularised to β_0^2/s_{eff} by interactions ($U > 0$, orange dashed), remaining finite everywhere.

in the BA through the channel splitting $2|s| \rightarrow 0$ at $\beta_0 = \gamma$. For $U > 0$ and $F(\lambda) > 0$, the interaction-corrected splitting $2s_{\text{eff}}$ remains finite and represents an avoided coalescence at the same bare control point (the R-L distance $|\mu^R - \mu^L| = 2\tilde{\Gamma}$ is nonzero everywhere and is not an EP diagnostic; see also Refs. [36, 37]). The dressed rapidities $\mu_{\pm}^{R/L}$ presented in Fig. 5 encode the full pseudo-Hermitian structure: the three-layer picture—

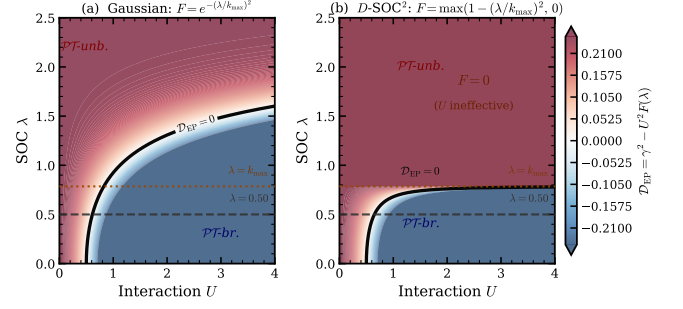


FIG. 6. EP-boundary diagnostic controlled by the SOC form factor. Color: $\mathcal{D}_{\text{EP}}(U, \lambda) = \gamma^2 - C_F U^2 F(\lambda)$, derived in the Supplemental Material in the adopted dimensionless normalization $C_F = 1$ and evaluated at the bare control point $\beta_0 = \gamma$. Positive (red): relative $\mathcal{PT}$ -unbroken ($\mathcal{D}_{\text{EP}} > 0$); negative (blue): relative $\mathcal{PT}$ -broken ($\mathcal{D}_{\text{EP}} < 0$). The thick black curve is the local EP-boundary diagnostic $\mathcal{D}_{\text{EP}}(U, \lambda) = 0$. Horizontal lines mark the SOC values used in Fig. 5: dashed $\lambda = 0.50$, dotted $\lambda = k_{\text{max}} = \pi/4$. **(a)** Gaussian form factor $F = e^{-(\lambda/k_{\text{max}})^2}$: interactions U regularise the EP for all λ ; the EP boundary curves to large U as $\lambda \rightarrow k_{\text{max}}$. **(b)** D-SOC² form factor $F = \max(1 - (\lambda/k_{\text{max}})^2, 0)$: at $\lambda = k_{\text{max}}$, $F = 0$ exactly (orange dotted line), so U cannot regularise the EP regardless of interaction strength—producing the sharp vertical wall in this analytic diagnostic. The figure illustrates the model dependence of the interaction-induced impurity-sector EP shift; it is not used as a standalone many-body phase diagram or an independent determination of the Kondo scale.

bare h_d eigenvalues, interaction-corrected $\tilde{\epsilon}_{\xi} \pm s_{\text{eff}}$, and Bethe-dressed $\mu_{\pm}^{R,U}$ —demonstrates how interactions replace the bare coalescence by an avoided one, with the channel splitting serving as the quantitative EP-distance diagnostic across all parameter regimes.

E. Frozen-model BA scattering kernel and screening-scale estimate

The frozen Bethe-Ansatz (BA) construction gives a channel-diagonal Breit-Wigner scattering amplitude. In what follows we use the logarithmic derivative of this frozen two-body scattering phase to define an integrability-based kernel and then match its dimensional scale to the same microscopic Anderson channel used in the Schrieffer-Wolff reduction. The result is therefore a BA-kernel screening-scale estimate for the frozen effective model, rather than a full thermodynamic Bethe-Ansatz determination of the Kondo temperature:

$$S_{\eta}(k) = \frac{k - \tilde{\epsilon}_{d,\eta} - i\tilde{\Gamma}_{\eta}}{k - \tilde{\epsilon}_{d,\eta} + i\tilde{\Gamma}_{\eta}}, \quad \eta = \pm. \quad (71)$$

The rapidity matching used below follows the usual frozen Anderson-Kondo scattering construction: the charge rapidity is measured from the dressed impurity level, $u = k - \tilde{\epsilon}_{d,\eta}$, and the logarithmic derivative of the

Breit–Wigner phase shift gives the kernel in Eq. (78). The phase-shift kernel fixes the scattering structure but does not by itself determine a unique thermodynamic energy scale. We therefore set the dimensionful scale by matching the same frozen channel to the Schrieffer–Wolff exchange. In this construction the imaginary parts and coalescence of the biorthogonal rapidities diagnose the EP structure; they are not identified directly with T_K . At the uniform particle-hole convention used in the final figures, $\epsilon_\xi = -U/2 = -1.0$ and $U = 2.0$, the two virtual charge denominators are both unity,

$$|\tilde{\epsilon}_d|^{-1} + |\tilde{\epsilon}_d + U|^{-1} = 1 + 1 = 2, \quad (72)$$

so the PH-symmetric exponent reduces to $-\pi/(4\chi_K \tilde{\Gamma}_\eta^{\text{bio}})$ in the dimensionless units of the frozen scale estimate. This is the rapidity-to-Anderson matching used for the SW/BA-kernel relative comparison of Supplemental Fig. S3.

For a reciprocal Hermitian Anderson channel,

$$\tilde{\Gamma}_\eta = \pi \rho_{b,\eta}(0) |V_\eta|^2, \quad (73)$$

while a non-Hermitian or nonreciprocal channel is described by the biorthogonal product

$$\tilde{\Gamma}_\eta^{\text{bio}} = \pi \rho_{b,\eta}(0) V_\eta^R V_\eta^L. \quad (74)$$

Only positive-real-part screening eigenvalues of the corresponding exchange matrix are interpreted as Kondo screening channels.

Spin-orbit coupling (SOC) enters the frozen scale through the active channel width,

$$\tilde{\Gamma}_\eta^{\text{bio}}(\lambda) = \pi \rho_{b,\eta}(0; \lambda) V_\eta^R(\lambda) V_\eta^L(\lambda). \quad (75)$$

The channel density of states contains the Rashba-split velocity Jacobian,

$$\rho_{b,\eta}(0; \lambda) = \sum_{k_*} \frac{\mathcal{J}_2(k_*)}{|\partial_k \epsilon_\eta(k; \lambda)|_{k=k_*}}, \quad \epsilon_\eta(k_*; \lambda) = 0, \quad (76)$$

where $\mathcal{J}_2(k) = k/(2\pi)$ in two dimensions. Unless the real part of the bath self-energy is explicitly retained, $\tilde{\epsilon}_{d,\eta}$ is kept as the frozen impurity level and the explicit SOC dependence is collected in $\tilde{\Gamma}_\eta^{\text{bio}}(\lambda)$. If such a real shift is retained, one replaces

$$\tilde{\epsilon}_{d,\eta} = \epsilon_d + \text{Re } \Sigma_\eta^R(0; \lambda). \quad (77)$$

With $u = k - \tilde{\epsilon}_{d,\eta}$, the logarithmic derivative of Eq. (71) gives the Lorentzian frozen BA-kernel,

$$\begin{aligned} K_\eta(u) &= \frac{1}{2\pi i} \frac{d}{du} \ln S_\eta(u) \\ &= \frac{1}{\pi} \frac{\tilde{\Gamma}_\eta}{u^2 + \tilde{\Gamma}_\eta^2}, \quad \int du K_\eta(u) = 1. \end{aligned} \quad (78)$$

This kernel fixes the frozen scattering phase. The associated screening-scale estimate follows only after matching

to the microscopic Anderson exchange; no separate identification of T_K with the imaginary weight of a Bethe string is required in the present frozen-kernel construction. The frozen Schrieffer–Wolff exchange is

$$J_\eta^{\text{SW}} = 2|V_\eta|^2 [|\tilde{\epsilon}_{d,\eta}|^{-1} + |\tilde{\epsilon}_{d,\eta} + U|^{-1}], \quad (79)$$

so that

$$T_{K,\eta}^{\text{BA}} \simeq D_{\text{uv},\eta} \exp \left[-\frac{\pi}{2\chi_K \tilde{\Gamma}_\eta (|\tilde{\epsilon}_{d,\eta}|^{-1} + |\tilde{\epsilon}_{d,\eta} + U|^{-1})} \right]. \quad (80)$$

In the SOC-resolved non-Hermitian channel one reads $\tilde{\Gamma}_\eta \rightarrow \tilde{\Gamma}_\eta^{\text{bio}}(\lambda)$. Thus SOC enhances or suppresses the frozen scale only by increasing or decreasing the positive screening-channel width, or by an explicitly included real level shift.

At particle-hole symmetry with no channel-dependent real shift, $\tilde{\epsilon}_{d,\eta} = -U/2$, Eq. (80) reduces to

$$T_{K,\eta}^{\text{BA}} \simeq D_{\text{uv},\eta} \exp \left[-\frac{\pi U}{8\chi_K \tilde{\Gamma}_\eta^{\text{bio}}(\lambda)} \right]. \quad (81)$$

If a real channel shift $\tilde{\epsilon}_{d,\eta} = -U/2 + \eta s_\epsilon$ is retained, the linear ηs_ϵ terms from the empty and doubly occupied virtual charge sectors cancel, giving for $|s_\epsilon| < U/2$

$$T_{K,\eta}^{\text{BA}} \simeq D_{\text{uv},\eta} \exp \left[-\frac{\pi(U^2 - 4s_\epsilon^2)}{8\chi_K \tilde{\Gamma}_\eta^{\text{bio}}(\lambda)U} \right]. \quad (82)$$

As $|s_\epsilon| \rightarrow U/2$, the local-moment Schrieffer–Wolff estimate breaks down and the system crosses into mixed-valence physics.

For a genuinely non-Hermitian channel, the scalar product $V_\eta^R V_\eta^L$ replaces $|V_\eta|^2$. A conventional screening scale is then attached only to positive-real-part eigenvalues,

$$T_K^{\text{BA,NH}} = D_{\text{uv}} \exp \left[-\frac{1}{\chi_K j_{\text{scr}}} \right], \quad (83)$$

$$j_{\text{scr}} = \max \{ \text{Re } j_\alpha : j_\alpha \in \text{spec}(\rho_b J^{\text{bio}}), \text{Re } j_\alpha > 0 \}. \quad (84)$$

A negative or complex hybridization product can produce a frozen non-Hermitian EP/crossover, but it is not by itself an enhanced thermodynamic Kondo temperature.

Near an impurity EP that feeds a positive screening channel, the biorthogonal residue may enhance the active width in the frozen estimate,

$$\tilde{\Gamma}_\eta^{\text{BA,EP}} = Z_\eta^{\text{bio}} \tilde{\Gamma}_\eta^{\text{bio}}(\lambda), \quad 1 \leq Z_\eta^{\text{bio}} \leq Z_{\text{max}}, \quad (85)$$

with the bounded diagnostic form

$$Z_\eta^{\text{bio}} = \min \left[Z_{\text{max}}, \frac{s_{\text{ref},\eta}}{\max(|s_{\text{eff}}|, s_{\text{floor}})} \right]. \quad (86)$$

Here $s_{\text{ref},\eta}$ is a smooth off-EP channel normalization, $s_{\text{floor}} > 0$ is the causal/numerical infrared floor, and Z_{max}

prevents a formal Jordan divergence from being interpreted as an observable. For the plotted analytic curve we use $s_{\text{ref},\eta} = 0.375$, $s_{\text{floor}} = 0.03$, and $Z_{\text{max}} = 25$. Writing the nonsingular projection onto the screening channel as $\mathcal{A}_\eta$, define

$$\mathcal{J}_\eta^{\text{eff}} = \mathcal{A}_\eta |V_\eta^R V_\eta^L| (|\tilde{\epsilon}_{d,\eta}|^{-1} + |\tilde{\epsilon}_{d,\eta} + U|^{-1}). \quad (87)$$

The corresponding frozen-model scale is

$$T_{K,\eta}^{\text{BA,EP}} \simeq D_{\text{uv},\eta} \exp \left[-\frac{\pi |s_{\text{eff}}|}{2\chi_K \mathcal{J}_\eta^{\text{eff}}} \right]. \quad (88)$$

Equation (88) is an analytic frozen-model scale, or EP-distance diagnostic, not the exact thermodynamic Kondo temperature of the full driven interacting problem.

X. NEAR-EP LOW-ENERGY SCALE DIAGNOSTICS

Near the EP, the condition number $\kappa_{\text{imp}} \sim C/|s_{\text{eff}}|$ provides a diagnostic of eigenvector nonorthogonality and of the associated redistribution of low-energy spectral weight within the SBMF treatment. In the conservative interpretation used here, Eq. (55) is a screening-eigenvalue estimate rather than a universal expression for the exact Kondo temperature. The low-energy response is phase-selective: it is most visible in the Kondo regime (Fig. 3) and is reduced both by large SOC, which suppresses the overlap factor $F(\lambda)$ in Supplemental Material, and by strong interaction-induced regularisation, which increases s_{eff} . The diagnostic mechanism is the near-collinearity of the right eigenvectors, $|r_+\rangle \approx |r_-\rangle$ as $s \rightarrow 0$, accompanied by the corresponding growth of the dual norm. Thus $\kappa_{\text{imp}} = \|R\| \|R^{-1}\|$ tracks the distance to the EP and the near-EP redistribution of impurity spectral weight, but it is not used as a standalone proof of a universal Kondo-scale enhancement. A proper Kondo-temperature enhancement is claimed only when the EP-active sector feeds a positive antiferromagnetic screening eigenvalue of $\rho_b J^{\text{bio}}$. The enhancement is exponential: SOC does not add a separate factor to T_K , but changes the screening width appearing in the denominator of the Kondo exponent. If the Rashba-split bath increases the active channel density of states or the chiral hybridization product $V_\eta^R V_\eta^L$, then $\tilde{\Gamma}_\eta^{\text{bio}}(\lambda)$ increases and the negative exponent in $T_{K,\eta}^{\text{BA,EP}}$ is reduced in magnitude. Conversely, if SOC lowers the active density of states or suppresses the form factor, the same expression suppresses the frozen Kondo scale. Near an impurity EP feeding a positive screening channel, the biorthogonal residue further renormalizes the width as $\tilde{\Gamma}_\eta^{\text{BA,EP}} = Z_\eta^{\text{bio}} \tilde{\Gamma}_\eta^{\text{bio}}$, where Z_η^{bio} is the bounded diagnostic factor of Eqs. (85)–(86). This gives an additional EP-assisted increase of the frozen scale as the regularized EP-distance is reduced, but not a literal divergent thermodynamic Kondo temperature.

XI. CONCLUSIONS

We have shown that a strictly Hermitian, periodically driven quantum impurity coupled to a spin-orbit-split Dirac bath develops emergent passive non-Hermitian behavior through a transparent and controllable microscopic mechanism. Bilinear Gaussian coarse graining over off-shell angular-momentum harmonics produces a real spin-odd coherent shift. Embedding this shifted channel in the energy-dependent retarded Dirac bath converts it into a spin-dependent width. After subtracting the common bath damping, the compensated relative kernel contains the passive $\mathcal{PT}$ core of Eq. (31), while the Kramers–Kronig partner $\Delta_{\text{coh}}\sigma_z$ acts as a causal detuning floor. The gauge-invariant amplitude $\tilde{\beta} = |\beta_0||b_c|$ controls the projected channel hybridization.

Four central and physically distinct results emerge:

First, the Hadamard rotation U_2 diagonalizes the hybridization block exactly, producing $\tilde{d}_\pm$ channels that couple exclusively to the $c_{k\pm}$ bath modes [Eq. (40)]. This diagonal structure makes the two-body S -matrix block-diagonal ($S^{LR} = S^{RL} = 0$), which is the origin of the channel-separated scattering used in the frozen-model Bethe-Ansatz construction.

Second, the condition number $\kappa_{\text{imp}} \sim C/|s_{\text{eff}}|$ shows a phase-selective EP response within the SBMF treatment, with the largest response in the Kondo regime and much weaker responses in the resonant-level and free-orbital regimes (Fig. 3). The interaction U renormalizes the EP distance s_{eff} and keeps the condition number finite. Thus κ_{imp} is used as a diagnostic of nonnormality and of the associated low-energy scale diagnostics, rather than as a standalone proof of a universal Kondo scale. The sign difference of the passive relative gain–loss channel alone does not split the Kondo scale, since the naive channel estimate depends on the squared detuning. Distinct channel scale estimates arise only when the SOC-split bath or the self-consistent saddle produces unequal channel widths or real detunings, as detailed in the Supplemental Material. The condition-number peaks in Figs. 1–3 are therefore interpreted as finite- $|\beta_0|$ EP diagnostics whose response is strongest in the Kondo regime, rather than as a separate universal “ $\tilde{\beta} \sim T_K$ ” precursor criterion.

Third, SOC enters the screening-scale estimate through the channel-resolved density of states, cutoff, chiral hybridization form factor, and one-body level shift. The frozen BA-kernel expression $T_{K,\eta}^{\text{BA}}(\lambda)$ should therefore be read as an integrability-based low-energy diagnostic, matched to the Schrieffer–Wolff channel, rather than as an independent exact thermodynamic identity. Its variation captures a combined SOC-projection and impurity-sector nonnormality effect within the frozen effective model, not a universal monotonic enhancement of T_K with SOC.

The location of the exceptional point is also important for this interpretation. In the mechanism studied here the relevant EP is localized in the effective impurity block and can modify the same positive screening channel

that enters the Schrieffer–Wolff exchange. In that case an enhanced biorthogonal residue may increase the corresponding frozen screening-scale estimate. An EP generated predominantly by the bath dispersion, reservoir self-energy, or an extended bath sector need not feed the local exchange in the same way. Such a bath-induced EP may alter spectral, transport, or decay properties without producing a comparable Kondo-scale enhancement. The connection between an EP and screening is therefore conditional on how the EP projects onto the low-energy impurity screening channel.

Fourth, the driven impurity exhibits a shifted Friedel-type resonance: the low-energy peak is referenced by $\omega_{\text{res}} \simeq \text{Re}(\tilde{\epsilon}_\xi) \neq 0$ [Eq. (54)]. This shift is expressed in terms of the gauge-invariant amplitude $|b_c|$, the spin-selective self-energy, and the renormalized impurity level; it is not attributed to the phase of the auxiliary slave-boson field. For the frozen kernel plotted in Fig. 4, the reference position is $\omega_{\text{res}} \simeq \text{Re}(\tilde{\epsilon}_\xi) = -1.0000$ at $\epsilon_\xi = -U/2 = -1.0$. A separate self-consistent finite-temperature SBMF closure may shift this value; such a shift should be quoted only when the same closure is used to generate the plotted spectrum.

Several directions open from these results. The biorthogonal Bethe-Ansatz equations [Eqs. (69)–(70)] provide an analytic frozen-model description, and the dressed rapidities in Fig. 5 reveal the essential pseudo-Hermitian structure: the two BA rapidities $\mu_\pm^{\text{BA}}$ encode the EP-distance structure in rapidity space. The bare $U = 0$ splitting vanishes continuously at $\beta_0 = \gamma$, whereas the interaction-corrected $2s_{\text{eff}}$ remains finite for $U > 0$ and $F(\lambda) > 0$, producing avoided rapidity coalescence. The bare coalescence suggests, by analogy with Jordan-block branch points, that the frozen effective model may develop singular low-energy density-of-states corrections near the merged peak. We do not identify the imaginary rapidity weight or this possible singularity directly with the Kondo temperature, and neither is used as an input to the scale estimate. Establishing any thermodynamic consequence beyond the frozen scattering diagnostic is left for future work. The SOC-dependent factor $F(\lambda) = \max(1 - \lambda^2/k_{\text{max}}^2, 0)$ introduces a sharp diagnostic boundary at $\lambda = k_{\text{max}}$ where this particular interaction correction to the impurity-sector EP vanishes, producing the vertical wall in Fig. 6. The shifted

channel-selective resonance at $\omega_{\text{res}} \neq 0$, together with relative gain–loss side features at scales set by $|\Gamma_{\text{PT}}|$, provides possible signatures in spin-resolved tunneling spectroscopy and non-equilibrium noise measurements in driven quantum dot devices. More broadly, our results support the comparatively narrow conclusion that an impurity-localized pseudo-Hermitian EP can enhance a low-energy screening diagnostic when it couples to a positive Kondo screening channel. They do not imply that every bath- or reservoir-induced EP enhances the Kondo scale.

SUPPLEMENTAL MATERIAL

Detailed derivations of the auxiliary-mode influence functional, Keldysh mean-field equations, physical-FDR construction, frozen SW/BA-kernel estimates, HWHM validity mask, negative signed-weight diagnostics, and parameter tables are provided in the Supplemental Material. The main text is kept self-contained by stating the effective projected kernel, the physical spectrum, the screening-scale estimates, and the SBMF spectral-weight transfer diagnostic explicitly.

CODE AND DATA AVAILABILITY

The numerical scripts and tabulated data used to generate the main and Supplemental figures are included in the accompanying reproducibility archive. The same versioned code and data are prepared for public release in the GitHub repository **emergent-pt-dirac-impurity** [38]; the repository will be activated when this revision is submitted. The archived files include the gauge-fixed frozen effective-model solver, the cleaned BA-kernel figure script, and the full Fig. 3 drive-range data referred to in the caption.

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